



LES-Based Investigation of Unsteady Wake Effects on the Rear Wing in Tandem WIG Configuration

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Abstract

Tandem Wing-in-Ground (WIG) configurations are inherently sensitive to wake–wing interactions, where the rear wing operates within the unsteady wake of the front wing. This results in time-varying aerodynamic loading that can adversely affect longitudinal stability and is not adequately captured by conventional steady-state analysis. The primary challenge lies in characterizing these unsteady effects to assist on design approaches that can mitigate wake-induced disturbances on the rear wing.

To address this, a computational framework combining steady Reynold Averaged Navier Stokes (RANS) and Large Eddy Simulation (LES) is employed. RANS simulations establish baseline aerodynamic performance and load distribution, while LES resolves the transient wake structures and their influence on rear-wing loading. Time-resolved lift and drag signals are analysed using spectral analysis methods, including Fourier transform and Welch’s method, to identify dominant frequencies and energy distribution within the wake-induced fluctuations.

Additionally, passive wake control is explored through the implementation of Gurney flaps on the front-wing endplates to modify wake structure and turbulence characteristics.

The results show that the rear wing is subjected to coherent, periodic aerodynamic forcing, with dominant frequency components indicating structured wake interaction rather than random turbulence. The introduction of Gurney flaps redistributes energy toward higher frequencies, reducing large-scale coherence but increasing overall unsteady energy and drag. This demonstrates that while wake homogenization can be improved, it comes at the expense of aerodynamic efficiency and does not eliminate unsteady loading.

Overall, the study concludes that unsteady wake effects are a governing factor in tandem WIG aerodynamics and must be explicitly resolved in both analysis and design. Passive flow control methods alone may be insufficient to stabilize the rear-wing environment, and future designs should integrate geometric optimization with control strategies that are tuned to the frequency characteristics of wake-induced disturbances.

Abbreviation

WIG	Wing-In-Ground
OGE	Out of Ground Effect
IGE	In Ground Effect
CFD	Computational Fluid Dynamic
LES	Large Eddy Simulation
RANS	Reynolds Averaged Navier Stokes
SGS	Subgrid-scale
FFT	Fast Fourier Transform
DFT	Discrete Fourier Transform
PSD	Power Spectral Density

Nomenclature

c	Wing chord length	m
b	Wingspan	m
h	Wing height at quarter-chord	m
h_{AG}	Air gap beneath endplate	m
C_L	Lift Coefficient	Non-dimensional
C_D	Drag Coefficient	Non-dimensional
C_{D0}	Profile Drag Coefficient	Non-dimensional
C_{Di}	Induced Drag Coefficient	Non-dimensional
AR	Wing Aspect Ratio	Non-dimensional
e	Oswald Efficiency Factor	Non-dimensional
α	Angle of Attack	degree
u_τ	Friction Velocity	m s ⁻¹
y_p	First Inflation Layer Height	m

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1. Introduction

The FWISH project is a compact, single-seat ground-effect vehicle prototype designed using a tandem wing configuration to achieve both lift generation and longitudinal stability within low-altitude flight regimes where ground-effect aerodynamics significantly influence overall performance.

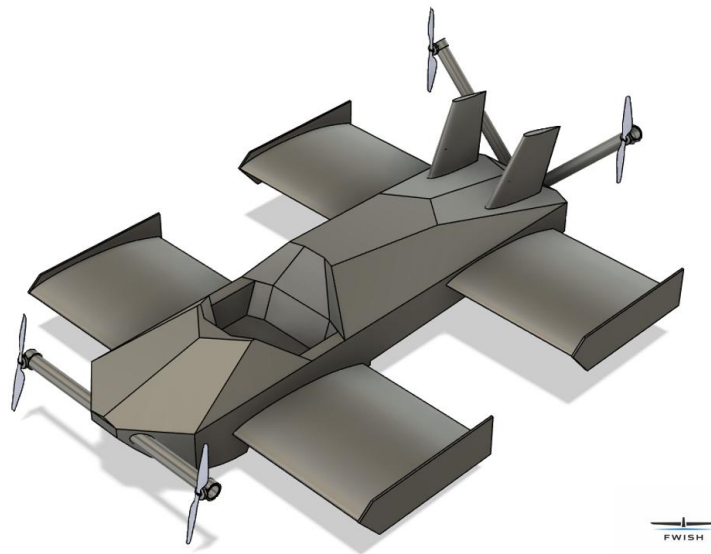


Figure 1: FWISH Prototype Model

In ground-effect operation, aerodynamic characteristics differ significantly from conventional flight due to the proximity of the lifting surfaces to the ground. This results in increased lift and reduced induced drag, primarily caused by the suppression of wingtip vortices and modification of the pressure distribution beneath the wing.

The tandem wing configuration adopted in FWISH distributes lift between a forward and rear lifting surface. However, this configuration introduces strong aerodynamic coupling between the two wings. In particular, the wake generated by the front wing interacts directly with the rear wing, altering its effective inflow conditions.

As a result, the rear wing does not operate in a uniform freestream, but within a disturbed flow field characterized by wake-induced velocity deficits, downwash, and vortex structures. These effects influence both the magnitude and temporal variation of aerodynamic loading, making the behaviour of the rear wing fundamentally different from that of an isolated wing.

1.1. Problem Statement

While the aerodynamic behaviour of isolated wings in ground effect is relatively well understood through existing empirical and theoretical studies, there is limited data available for tandem configurations where wake interaction between lifting surfaces is present. In particular, the influence of front-wing-generated vortices and wake structures on the aerodynamic loading of the rear wing remains insufficiently characterized.

This lack of understanding limits the ability to accurately predict aerodynamic performance and stability behaviour in tandem ground-effect configurations.

1.2. Objective

The objective of this study is to investigate the aerodynamic behaviour of a tandem wing ground-effect configuration representative of the FWISH design, with particular focus on the wake interaction between the front and rear wings during level flight.

Computational Fluid Dynamics (CFD) is employed to characterise the flow field and aerodynamic loading, beginning with a baseline analysis of an isolated wing in ground effect, followed by the tandem configuration to assess the influence of front-wing-generated wake on the rear wing.

The study focuses on identifying wake-induced flow structures and their impact on rear-wing performance, with the aim of informing design strategies to improve overall stability.

1.3. Theoretical Framework for Induced Drag in Ground Effect

The theoretical foundation of wing aerodynamics is based on the lifting-line theory developed by Lanchester and Prandtl, which established that lift and drag are governed by circulation around the wing and the formation of trailing vortices. For a finite wing, the total drag coefficient can be expressed as the sum of profile drag and induced drag:

$$C_D = C_{D0} + \frac{C_L^2}{\pi A R e} \quad (1)$$

Where C_{D0} represents the profile drag coefficient, AR is the aspect ratio, e is the Oswald efficiency factor. This relationship is commonly written in a simplified quadratic form:

$$C_D = C_{D0} + kC_L^2 \quad (2)$$

The second term represents induced drag, which arises from the downwash generated by trailing vortices and is directly related to lift production. Prandtl and Wieselsberger further developed this to account for ground proximity, introducing an influence coefficient to modify the induced drag component:

$$C_{Di,IGE} = (1 - \sigma) \frac{C_L^2}{\pi AR} \quad (3)$$

The term $(1 - \sigma)$ therefore represents the ratio of induced drag in ground effect (IGE) to that out of ground effect (OGE) or freestream conditions. As the wing operates closer to the ground, σ increases, resulting in a reduction of the induced drag coefficient [1, 2].

Modern research builds upon these classical theories to provide more precise assessments of geometry. Ashill extended this framework by incorporating empirical observations for wings operating in ground effect with endplates, quantifying the additional reduction in induced drag beyond that predicted by ground effect alone [1, 2].

The induced drag reduction for a wing with endplate (WEP) is expressed as:

$$(1 - \sigma)_{WEP} = 1.5 \left(\frac{h_{AG}}{b} \right) + 5 \left(\frac{h}{b} \right) - \frac{10}{\left(\frac{h_{AG}}{b} \right)^{\frac{1}{5}}} \left(\frac{h}{b} \right)^2 \quad (4)$$

$$\frac{h}{b} = \frac{h_{TE}}{b} + \frac{3}{4} \frac{\sin \alpha}{AR} \quad (5)$$

where h_{AG} is the air gap beneath the endplate, h_{TE} is the wing height at the trailing edge, h is the wing height at the quarter-chord, and b is the wingspan. This formulation accounts for the combined influence of ground proximity and endplate geometry on induced drag reduction and will be used as a reference for comparison with the CFD results in subsequent analysis.

1.4. RANS and LES for Flow Modelling

Accurate prediction of wake-induced aerodynamic behaviour requires appropriate turbulence modelling strategies. This study incorporates both Reynolds-Averaged Navier–Stokes (RANS)

and Large Eddy Simulation (LES) to investigate the aerodynamic behaviour of the tandem wing configuration.

While RANS methods are effective in predicting mean aerodynamic quantities, they are limited in resolving unsteady vortex structures in wake-dominated flows. LES addresses this limitation by directly resolving large, energy-containing turbulent eddies while modelling smaller subgrid-scale (SGS) motions through filtering of the Navier–Stokes equations. This enables accurate capture of transient flow features such as vortex shedding, wake interaction, and flow separation, providing a more representative description of real flow conditions [3].

LES is particularly suitable for the present study, where the rear wing operates within the unsteady wake of the front wing under ground effect. The resulting time-dependent aerodynamic loading cannot be captured using steady RANS alone. However, due to its higher computational cost, RANS is first used to establish baseline flow characteristics and ensure mesh adequacy before transitioning to LES for detailed unsteady analysis.

1.5. Design Specifications

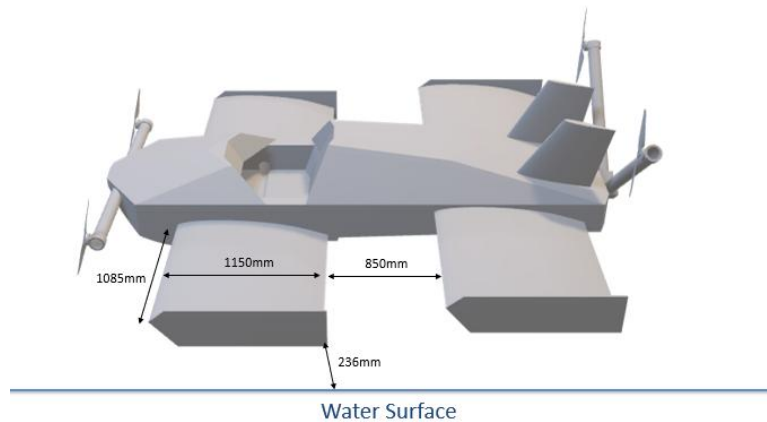


Figure 2: FWISH Baseline Geometry and Operating Altitude

FWISH configuration is defined as shown in Figure 2. The vehicle's wing chord, span, longitudinal spacing between the front and rear wings, and height of airgap beneath the endplate are established based on this baseline design.

The vehicle is designed to cruise at a target freestream velocity of around 25 m/s, which is adopted as the reference condition for the present study.

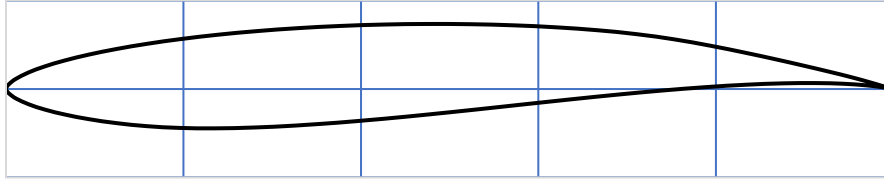


Figure 3: Eppler 817 Airfoil



Figure 4: Wing Section and Endplate Drawing

The wing section is based on the Eppler 817 airfoil, originally developed by Richard Eppler for low-speed aerodynamic applications [5]. The airfoil is selected for its favourable performance characteristics, exhibiting a relatively high camber and thickness distribution that enables efficient lift generation at moderate angles of attack while maintaining stable pressure recovery. In the baseline design, the wing is mounted at a geometric angle of attack of 5 degree relative to the horizontal reference plane of the vehicle and is equipped with vertical endplates at the tips, as illustrated in Figure 4.

2. Methodology

The analysis considers the lifting surfaces only, with the configuration modelled as a tandem wing system without inclusion of the fuselage or ancillary components to isolate the aerodynamic effects of ground interaction and inter-wing wake coupling. Hence, for this study, value of b is doubled.

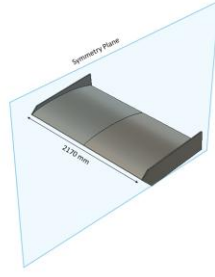


Figure 5: Computational Model of the Wing Configuration

To reduce computational cost, a half-span model is employed with a symmetry boundary condition imposed along the longitudinal centre plane, representing a full-span configuration under the assumption of symmetric flow.

A staged computational approach is adopted, beginning with steady RANS simulations and subsequently extended to LES. The initial step involves simulating an isolated wing in ground effect to establish baseline aerodynamic characteristics. The computed lift and drag coefficients are compared against available empirical and theoretical relationships to verify the numerical setup, including mesh, boundary conditions, and turbulence modelling.

Following this validation, the same setup is applied to the tandem wing configuration by introducing the rear wing at the prescribed longitudinal spacing. The RANS simulations provide the mean aerodynamic behaviour of the coupled system, forming a reference for further analysis. The study is then extended to LES to obtain time-resolved flow field data, enabling detailed investigation of wake-induced aerodynamic loading on the rear wing.

An additional configuration incorporating an endplate-mounted Gurney flap is also considered to assess its influence on wake structure and downstream aerodynamic behaviour under identical simulation conditions.

2.1. Computational Setup

The computational framework is divided into two distinct setups corresponding to the RANS and LES analyses. The RANS setup is established to obtain baseline aerodynamic characteristics and to verify the numerical configuration, while the LES setup is developed to resolve the unsteady flow structures associated with wake interaction.

ANSYS Fluent 2021 R2 is employed for all the simulations. The RANS simulations are conducted using coupled pressure–velocity scheme to ensure robust convergence of the mean flow solution. For the LES simulations, PISO scheme is employed to improve computational efficiency in resolving unsteady flow structures.

2.2. RANS Simulation and Validation

The RANS simulations are conducted using two domain configurations to evaluate both free-flight and ground-effect conditions at $Re = 1,970,000$.

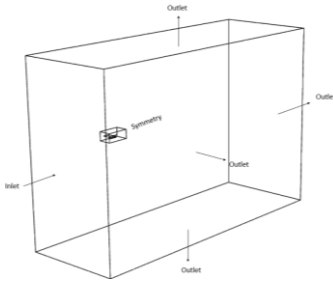


Figure 6: Domain Configuration –
OGE

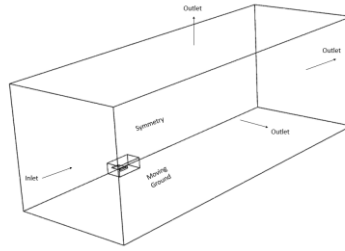


Figure 7: Domain Configuration –
IGE

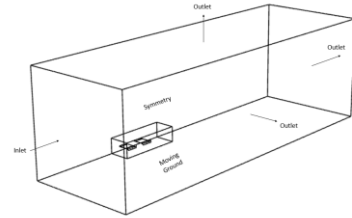


Figure 8: Domain Configuration –
Tandem Wing IGE

The free-flight configuration is first established to validate the aerodynamic performance of the wing, including profile and induced drag, as illustrated in Figure 5.

The ground-effect configuration is subsequently introduced, as shown in Figure 6, by replacing the lower boundary with a moving wall at freestream velocity while maintaining consistent domain extents. The distance between the wing and the ground is systematically varied to investigate its influence on the aerodynamic behaviour. This enables direct comparison of flow characteristics with and without ground influence.

Following validation of the isolated wing case, the setup is extended to the tandem wing configuration, as illustrated in Figure 7, by introducing the rear wing at the prescribed longitudinal spacing while maintaining identical numerical conditions.

In all configurations, mesh refinement is applied near the wing surface and wake region, with near-wall resolution maintained for appropriate y^+ values. The simulations are performed using

a steady-state solver with the $k-\omega$ SST turbulence model. The computational domain is defined using a near-field and far-field region to capture both local flow features and far-field effects. The near-field extends 1 m upstream and 2 m downstream of the wing, while the far-field extends 10 m upstream and 20 m downstream. For the free-flight configuration, the vertical domain extent is defined symmetrically about the wing to minimise boundary influence. In contrast, for the ground-effect configuration, the vertical extent of both near-field and far-field regions is defined by h_{AG} .

2.2.1. Mesh Convergence

To ensure numerical accuracy, a mesh convergence study is performed using the isolated wing configuration under ground-effect conditions at a representative height of $h_{AG}/c = 0.3113$ selected as the midpoint of the height-to-chord ratios investigated ($h_{AG}/c = 0.1374 - 0.6591$). This choice ensures that the selected case is representative of the overall flow behaviour across the ground-effect regime.

Three mesh levels generated by systematically varying mesh parameters, including local element sizing in the wing region, near-field and far-field domains, as well as the boundary layer resolution at the wing and ground surfaces.

	Cells	Faces	Nodes
Coarse	1,655,265	7,577,478	4,454,396
Medium	2,838,387	12,871,773	7,538,437
Fine	4,677,357	21,464,868	12,680,993

Table 1: Mesh

	Wing Local Sizing [m]	Near Domain Local Sizing [m]	Far Domain Local Sizing [m]	Wing Boundary Layer	Ground Boundary Layer
Coarse	0.05	0.1	2	20	20
Medium	0.025	0.05	1	25	25
Fine	0.0125	0.025	0.5	30	30

Table 2: Mesh Settings

The assessment of mesh adequacy is based on multiple criteria. Near-wall resolution is evaluated through the dimensionless wall distance y^+ as defined below:

$$y^+ = \frac{\rho y_p u_\tau}{\mu} \quad (6)$$

Where u_τ is the friction velocity, μ is the viscosity of the fluid, and y_p is the height of the first cell of the inflation layer [6, 7]. This distribution is maintained below 1 to ensure proper resolution of the viscous sublayer for the selected turbulence model.

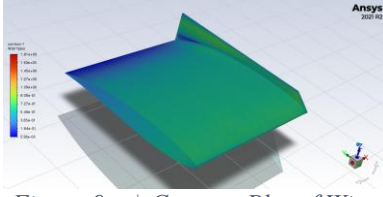


Figure 9: y^+ Contour Plot of Wing Top Surface – Medium Mesh

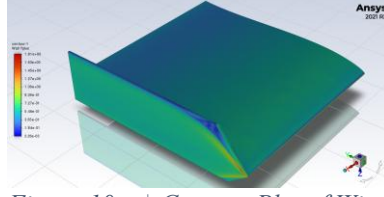


Figure 10: y^+ Contour Plot of Wing Bottom Surface – Medium Mesh

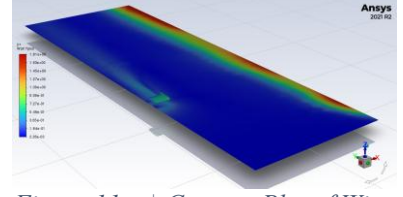


Figure 11: y^+ Contour Plot of Wing and Ground – Medium Mesh

The complete y^+ contour plots are provided in the Appendix A for reference.

Global conservation is also verified by monitoring the net mass flow rate across the domain. In addition, the stability of the solution is assessed through the convergence behaviour and fluctuation of aerodynamic coefficients after stabilisation.

	Net Mass Flow Rate [%]	C_L Fluctuation	C_D Fluctuation
Coarse	-0.0041%	1.14%	2.49%
Medium	-0.0055%	0.18%	0.40%
Fine	0.0052%	0.45%	0.59%

Table 3: Global Conservation Monitoring Result

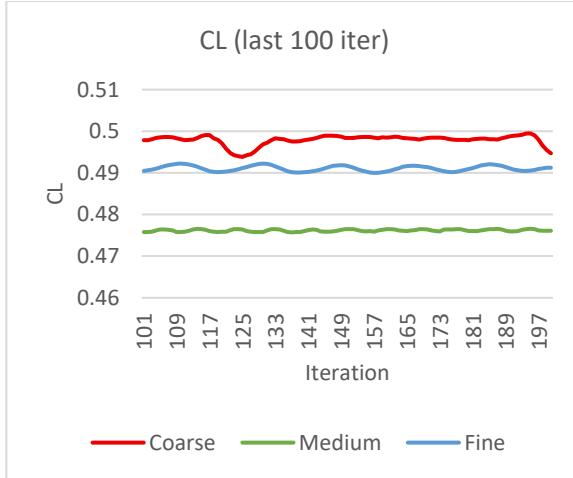


Figure 12: Lift Coefficient Plot of Last 100 Iterations

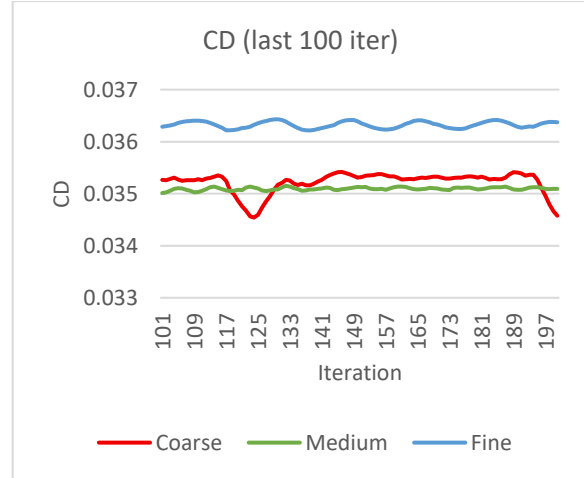


Figure 13: Drag Coefficient Plot of Last 100 Iterations

These criteria are used across mesh levels to determine grid independence and selected the medium mesh as the appropriate mesh for subsequent simulations.

2.2.2. Simulation Cases

A series of simulations are conducted to characterise the aerodynamic behaviour of the wing under free-flight and ground-effect conditions, as well as to assess the effect of geometric variation and tandem configuration.

	Parameter	Values
OGE	α	-5 to 5 degrees
IGE	h/c	0.13 to 0.66
Tandem Wing IGE	h/c	0.3113

Table 4: Steady RANS Simulation Cases

The free-flight configuration is simulated over a range of angles of attack to establish the baseline aerodynamic characteristics. Namely the profile drag and induce drag factor are determined using Equation (2).

Subsequently, the isolated wing in ground effect is simulated at multiple height-to-chord ratios and compared with empirical values from Equation (4).

Finally, the tandem wing configuration is also analysed at $h_{AG}/c = 0.3113$.

2.2.3. Out of Ground Effect Validation and Aerodynamic Characteristics

α (deg)	Mean C_L	Mean C_D
5	0.395875316	0.037888241
3	0.29349593	0.027474774
0	0.143741898	0.01784336
-3	-0.01275413	0.015732492
-5	-0.117459561	0.01918819

Table 5: Single Wing OGE Simulation Results

The free-flight simulation results are validated against the classical drag formulation given by Equations (1) and (2). The variation of lift coefficient with angle of attack, shown in Figure 26, exhibits a linear relationship consistent with lifting-line theory in the linear regime. The corresponding drag behaviour in Figure 27 follows the expected quadratic trend.

The drag curve exhibits a local minimum, corresponding to the condition $\frac{dC_D}{d\alpha} = 0$, indicating the minimum total drag state. From the simulation results, the minimum drag coefficient is $C_{D,min} \approx$

0.0157, where the induced drag contribution is minimised and the total drag approaches the profile drag component.

Further validation is performed by plotting C_D against C_L^2 as shown in Figure 28. A close to linear relationship is observed, where the slope corresponds to the induced drag factor k . From the linear fit, $k \approx 0.1387$ is obtained.

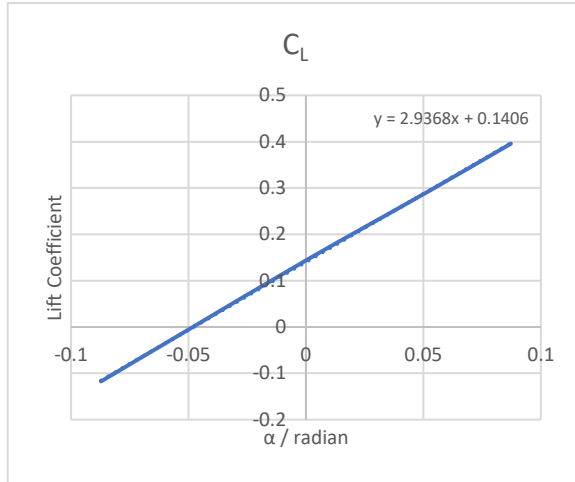


Figure 14: C_L vs α – Single Wing OGE

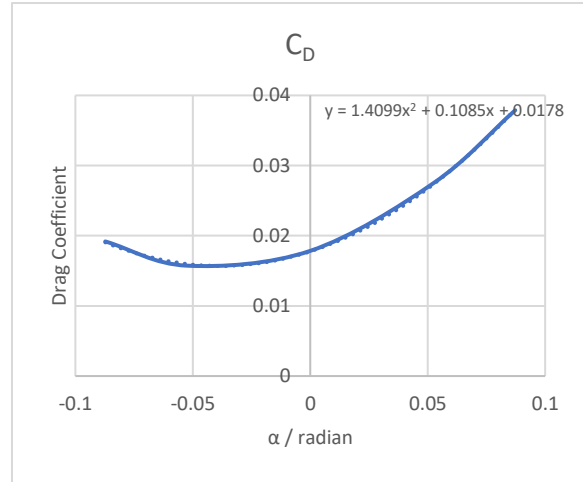


Figure 15: C_D vs α – Single Wing OGE

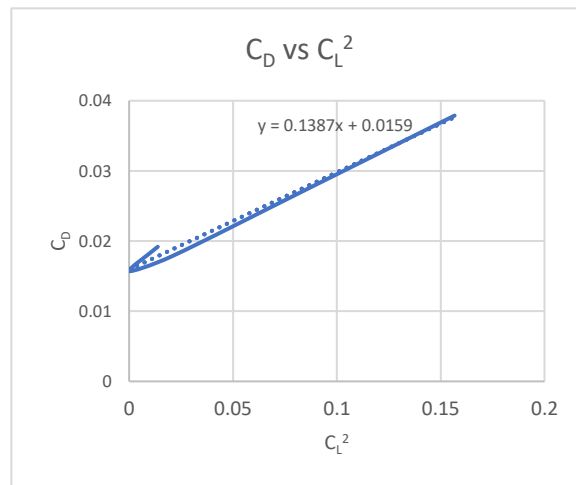


Figure 16: C_D vs C_L^2 – Single Wing OGE

2.2.3.1. Aspect Ratio Correction for Endplate Effects

To account for the presence of endplates, the effective aspect ratio is evaluated using Hoerner's correlation [4]. The effective aspect ratio is expressed as:

$$AR_{eff} = AR \left[1 + 1.9 \left(\frac{h}{b} \right) \right] \quad (7)$$

For the present configuration, the baseline aspect ratio is $AR = 1.89$, which increases to an effective aspect ratio of $AR_{eff} = 2.40$ based on the corresponding endplate height-to-span ratio. Using the induced drag factor k obtained from the quadratic drag polar, the Oswald efficiency factor is evaluated as:

$$e = \frac{1}{\pi AR k} \quad (8)$$

Substituting the corresponding values yields $e \approx 0.955$.

The induced drag coefficient C_{Di} is evaluated using both the baseline aspect ratio and the corrected effective aspect ratio. The total drag coefficient is then obtained by combining the induced drag with the profile drag component obtained as defined in Equation (1).

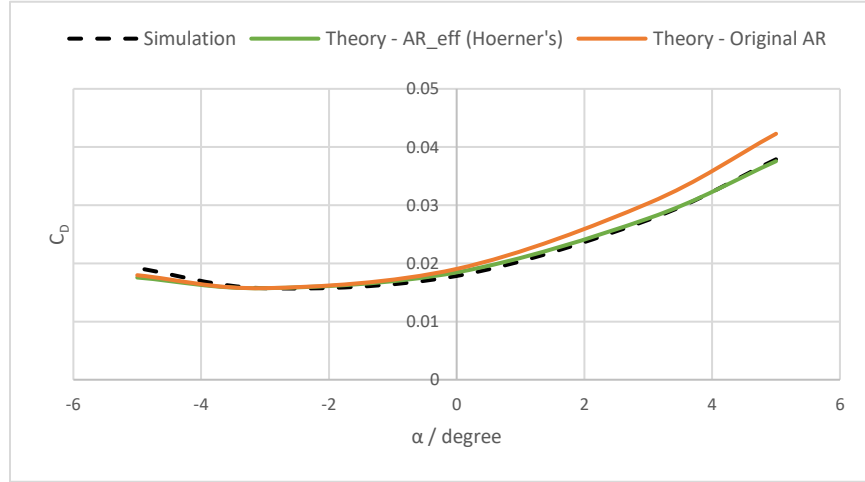


Figure 17: Total Drag Plot Comparison

α	Deviation – Original AR	Deviation – AR_{eff} (Hoerner's)
5	11.60%	-0.86%
3	10.39%	0.94%
0	6.75%	3.43%
-3	0.06%	0.03%
-5	-6.35%	-8.44%

Table 6: Total Drag Theory Deviation to Simulation

These results confirm that the CFD simulations capture both the linear lift behaviour and the quadratic drag characteristics of the wing in free-flight conditions, thereby validating the numerical setup for subsequent ground-effect and tandem wing analyses.

2.2.4. In Ground Effect Validation

h_{AG}/c	h/b	C_L	C_D
0.137391	0.107452455	0.650188	0.03042
0.18087	0.13049393	0.55481	0.032566
0.224348	0.153535405	0.513579	0.03398
0.267826	0.176576879	0.490674	0.034655
0.311304	0.199618354	0.476172	0.035095
0.398261	0.245701303	0.456343	0.035826
0.485217	0.291784253	0.44422	0.03617
0.65913	0.383950151	0.428947	0.036604

Table 7: IGE Simulation Results

To quantify the reduction in induced drag under ground effect, the induced drag correction factor is evaluated using the CFD results presented in Table 7. The formulation in Eq. (3) is rearranged such that the correction factor can be directly obtained from the simulated aerodynamic coefficients as

$$(1 - \sigma) = \frac{C_{D,sim} - C_{D0}}{C_L^2 / \pi AR} \quad (9)$$

where $C_{D,sim}$ is the total drag coefficient obtained from the CFD simulation in ground effect, C_{D0} is the profile drag coefficient determined from the out-of-ground-effect condition, and AR is the geometric aspect ratio of the wing.

These results are subsequently compared against Ashill's empirical formulation given in Eq. (4), which accounts for the combined effects of ground proximity and endplate geometry.

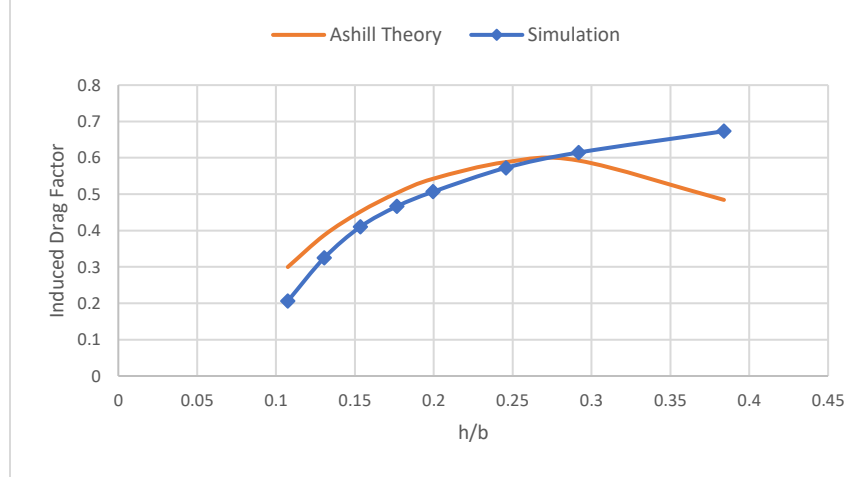


Figure 18: Simulation vs Empirical Plot of Induced Drag Factor

h_{AG}/c	h/b	$(1 - \sigma)_{Ashill}$	$(1 - \sigma)_{sim}$	Deviation
0.137391	0.107452455	0.299731143	0.206232219	-31.19%
0.18087	0.13049393	0.387035201	0.324567019	-16.14%
0.224348	0.153535405	0.452076955	0.410561014	-9.18%
0.267826	0.176576879	0.503247869	0.466401384	-7.32%
0.311304	0.199618354	0.542490374	0.506736089	-6.59%
0.398261	0.245701303	0.588141796	0.572540948	-2.65%
0.485217	0.291784253	0.592510148	0.614577588	3.72%
0.65913	0.383950151	0.484291648	0.673088356	38.98%

Table 8: Quantitative Comparison of Induced Drag Reduction Factor from CFD Simulation and Ashill's Empirical Prediction

The comparison between the CFD-derived $(1 - \sigma)$ and Ashill's empirical formulation shows noticeable deviation. The discrepancy at higher h/b values is expected, as Ashill's model is only valid up to $h/b \approx 0.27$. beyond which its predictive capability diminishes. At lower h/b , the deviation is attributed to differences in modelling conditions and near-ground flow physics. The present CFD employs a moving solid ground boundary, whereas Ashill's formulation is based on experimental conditions that may involve different surface characteristics that is not specified in detail. These difference influence boundary layer behaviour and vortex development.



Figure 19: Pressure Contour at $h/b = 0.15353405$



Figure 20: Pressure Contour at $h/b = 0.13049393$

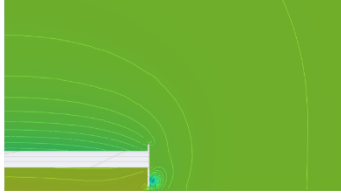


Figure 21: Pressure Contour at $h/b = 0.107452455$

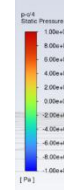


Figure 22: Pressure Contour Range

Furthermore, as the wing approaches the ground, strong pressure build-up occurs beneath the wing due to flow confinement. This is evident in the pressure contour plots, where a steep increase in pressure is observed for decreasing h/b . This pressure trapping effect enhances lift and suppresses induced drag more significantly than predicted by the empirical model, which does not fully capture this hike in near-ground effects.

Despite these deviations, the overall trend is well represented. Both the CFD results and Ashill's formulation show increasing induced drag reduction as the wing approaches the ground, indicating that the fundamental ground-effect behaviour is consistently captured.

2.3. LES Simulation

The LES simulations computational domain are enhanced adequately to capture the development and convection of unsteady wake structures. The domain configuration is extended relative to the RANS setup, with the near-field region increased to 2 m upstream and 8 m downstream of the wing. This modification provides sufficient spatial extent for the resolution of transient vortex structures and their downstream evolution.

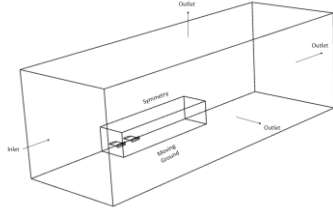


Figure 23: Domain Configuration for LES

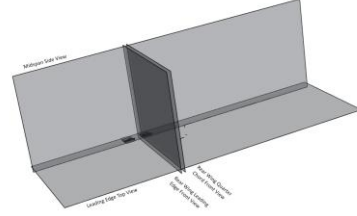


Figure 24: Analysis Planes

2.3.1. Mesh Refinement

Mesh refinement for the LES setup is guided by the Integral length scale estimated from the preceding RANS simulation. The local integral length scale is defined as:

$$l_0 = \frac{k^{1/2}}{C_\mu \omega} \quad (10)$$

where k is the turbulent kinetic energy, ω is the specific dissipation rate, and $C_\mu = 0.09$ is the model constant used in this study. This length scale represents the average size of the energy-containing eddies in the turbulent flow and provides a reference for determining the required grid resolution [8].

To assess the adequacy of the mesh for LES, the ratio between the turbulence length scale and the local grid size Δ is evaluated:

$$f = \frac{l_0}{\Delta} = \frac{k^{1/2}}{C_\mu \omega \cdot \Delta} \quad (11)$$

This parameter indicates the extent to which turbulent structures are resolved by the computational grid [8].

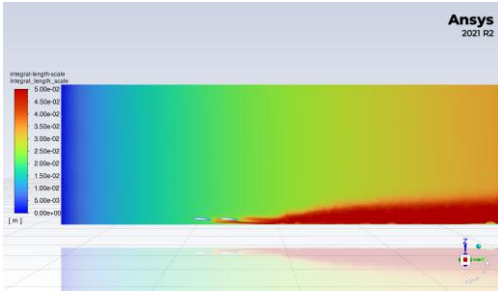


Figure 25: Integral Length Scale Contour – Midspan Side View

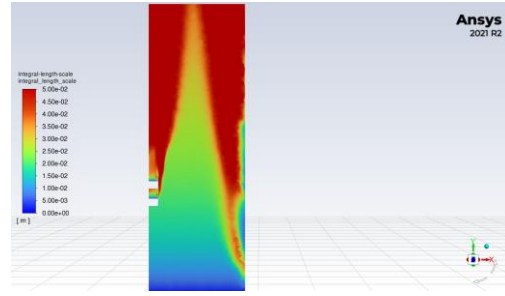


Figure 26: Integral Length Scale Contour – Leading Edge Top View

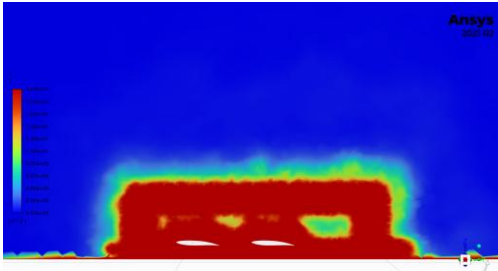


Figure 27: Resolution Parameter f (target $f = 2$) – Midspan Side View

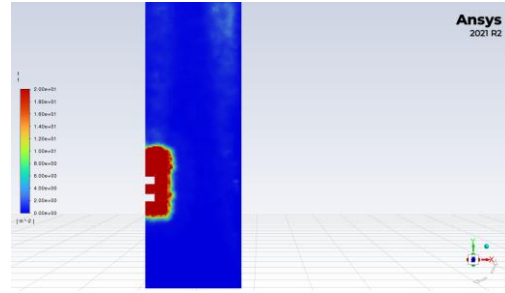


Figure 28: Resolution Parameter f (target $f = 2$) – Leading Edge Top View

The integral length scale contours shown in Figures 14 and 15 illustrate the distribution of turbulent structures within the flow field prior to mesh refinement. The formation of coherent vortex trails originating from the wingtip can be clearly observed, extending downstream into the wake region. This indicates the presence of dominant, energy-containing eddies associated with the tip vortices.

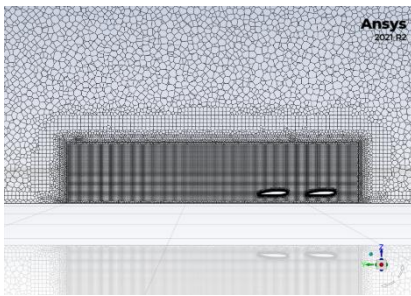


Figure 29: Refined LES Mesh – Midspan Side View

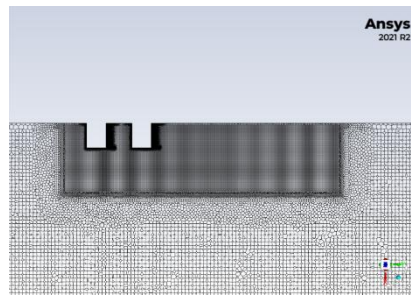


Figure 30: Refined LES Mesh – Leading Edge Top View

A refined mesh is adopted to improve spatial resolution for the LES simulations while maintaining computational efficiency resulting with refined mesh consists of 11,858,921 cells, providing sufficient resolution to capture the dominant turbulent structures in the flow. Further

refinement is guided by established recommendations of $f \geq 5$ that typically resolves approximately 80% of the turbulent kinetic energy [8].

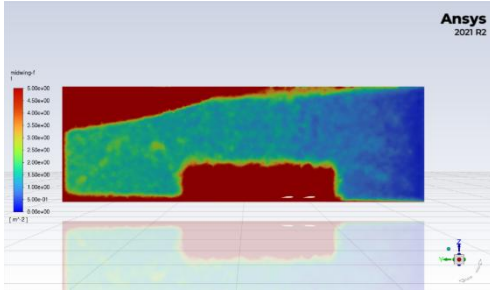


Figure 31: Resolution Parameter f (Refined Mesh, Target $f = 5$) – Midspan Side View

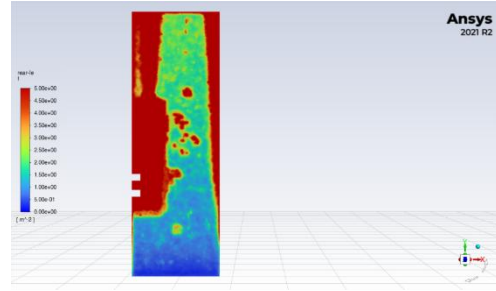


Figure 32: Resolution Parameter f (Refined Mesh, Target $f = 5$) – Leading Edge Top View

The LES simulations are conducted using a fixed time step of $\Delta t = 0.001s$ and are run for a total of 4000-time steps.

2.3.2. Geometric Modification for Wake Interaction Control

Following the baseline LES simulations, an additional geometric modification is introduced to investigate passive strategies for altering the wake structure impinging on the rear wing. The modification consists of a Gurney flap mounted on the endplate, with an extrusion height of 23 mm and a thickness of 10 mm.

The underlying hypothesis is that introducing asymmetry at the endplate will perturb the coherent vortex structures generated at the wingtip, causing a deviation in their trajectory and promoting earlier breakdown into smaller-scale turbulent structures. By reducing the coherence of the incoming wake, the flow field experienced by the rear wing is expected to become more isotropic, thereby potentially improve aerodynamic stability and reducing low-frequency load fluctuations.

Two configurations are considered, namely inward-facing and outward-facing Gurney flap orientations, as shown in Figures 33 and 34. These cases are analysed under identical numerical conditions as the baseline configuration to isolate the effect of vortex manipulation on the downstream flow.

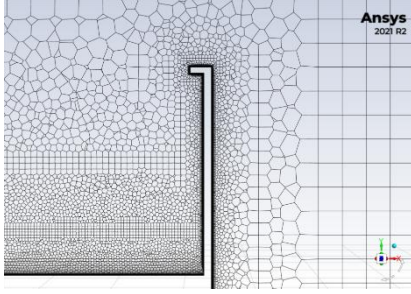


Figure 33: Gurney Flap Configuration – Inward Orientation

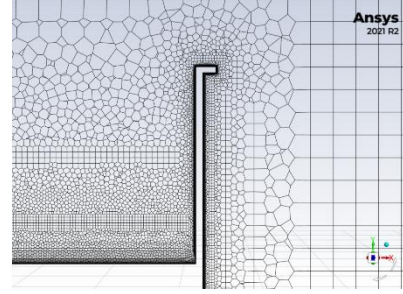


Figure 34: Gurney Flap Configuration – Outward Orientation

3. Results and Analysis

The aerodynamic behaviour of the tandem wing configuration is first assessed using steady RANS simulations to establish baseline characteristics. This is followed by LES to resolve unsteady wake structures and their influence on rear-wing loading.

3.1. Steady RANS Analysis

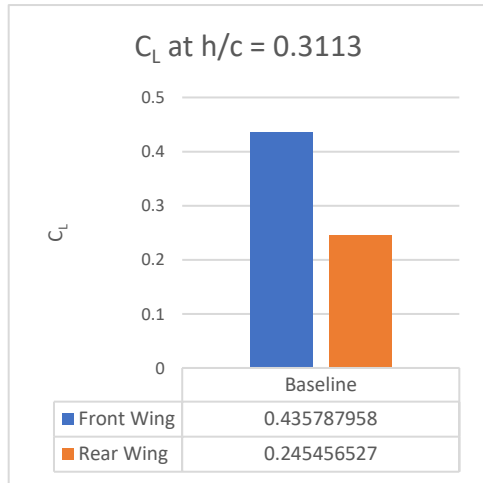


Figure 35: C_L Comparison of Front and Rear Wing

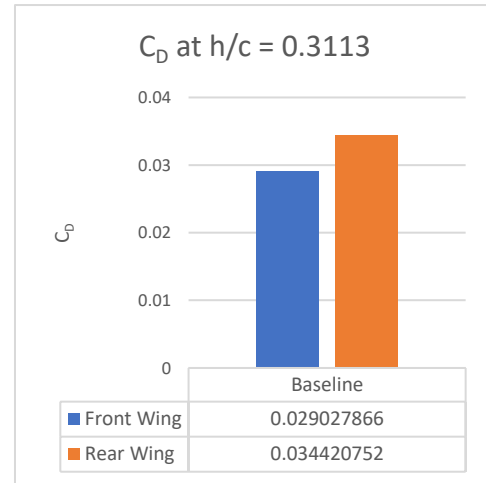


Figure 36: C_D Comparison of Front and Rear Wing

The steady RANS results show a significant redistribution of aerodynamic loading between the front and rear wings. At $h/c = 0.3113$, the rear wing experiences a reduction in lift of approximately 44% and an increase in drag of about 18% relative to the front wing.

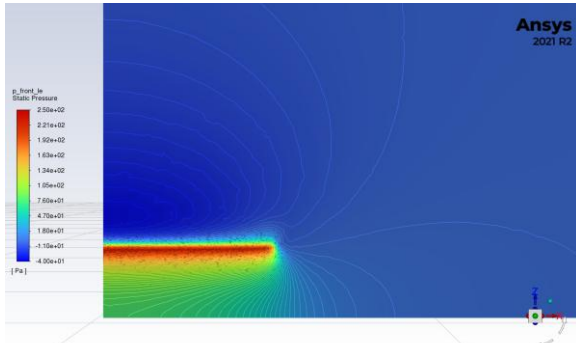


Figure 37: Pressure Contour – Front Wing Leading Edge Front View

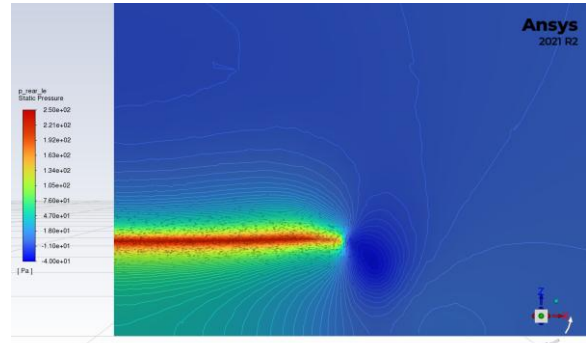


Figure 38: Pressure Contour – Rear Wing Leading Edge Front View

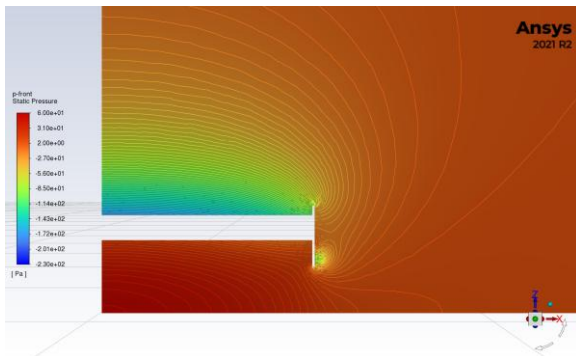


Figure 39: Pressure Contour – Front Wing Quarter Chord Front View

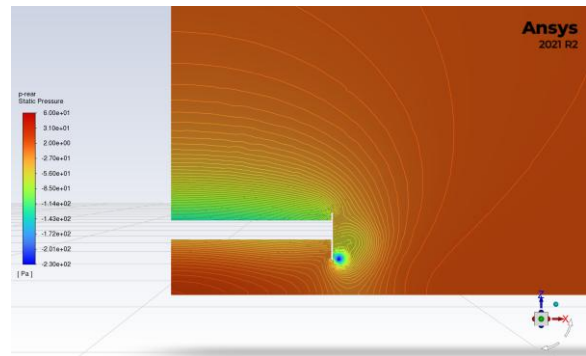


Figure 40: Pressure Contour – Rear Wing Quarter Chord Front View

This indicates that the rear wing operates in a degraded flow environment due to the wake of the front wing, resulting in reduced aerodynamic efficiency. As shown in Figures 37 and 38, the pressure contours at the leading edge reveal that the stagnation point of the rear wing is shifted slightly upward near the tip region, indicating an altered inflow direction caused by the upstream wake.

Furthermore, the quarter-chord pressure distribution (Figures 39 and 40) shows a weaker pressure gradient over the rear wing compared to the front wing, consistent with its reduced lift generation. At the same time, a stronger vortex structure is observed near the endplate of the rear wing, suggesting intensified local flow rotation and three-dimensional effects induced by wake interaction.

These observations establishes a baseline for subsequent unsteady LES analysis.

3.2. Unsteady LES Analysis

The LES results are trimmed to start from $t = 2s$ to exclude initial transient effects and ensure that only statistically developed flow is analysed.

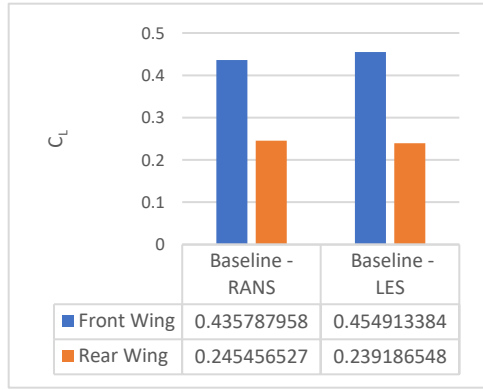


Figure 41: C_L Comparison for RANS and LES

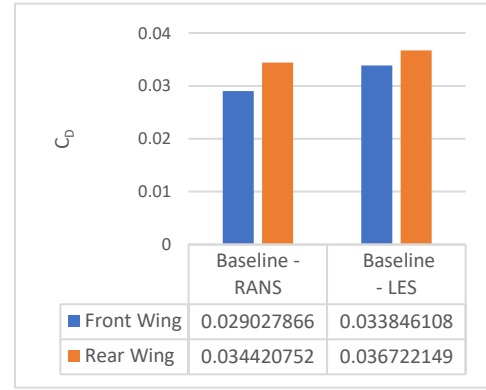


Figure 42: C_D Comparison for RANS and LES

Compared to the steady RANS results, the LES time averaged results show a slight increase in lift for the front wing of approximately 4%, while the rear wing experiences a reduction in lift of about 3%. In terms of drag, both wings exhibit an increase, with the front wing increasing by approximately 14% and the rear wing by around 6%.

These differences indicate that unsteady effects captured by LES lead to increased aerodynamic losses and further degradation of rear-wing performance under wake interaction.

The instantaneous vorticity contours reveal the presence of a dominant vortex structure near the tip of the rear wing. In the leading-edge front view (Figure 43), a large counterclockwise rotating vortex is observed, which alters the local inflow direction and contributes to the upward shift of the stagnation point previously identified in the steady RANS analysis.

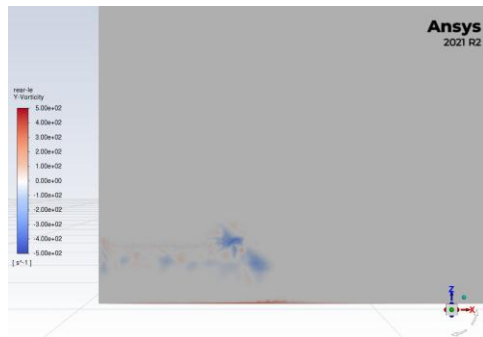


Figure 43: Instantaneous Vorticity Contour (Baseline) – Rear Wing Leading Edge Front View

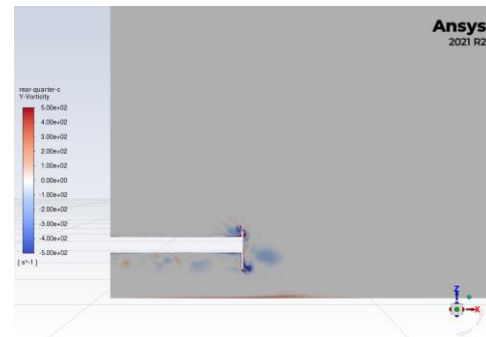


Figure 44: Instantaneous Vorticity Contour (Baseline) – Rear Wing Quarter Chord Front View

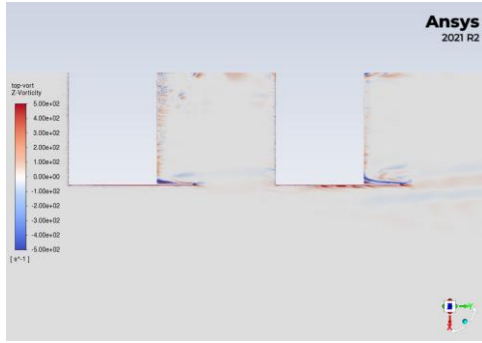


Figure 45: Instantaneous Vorticity Contour (Baseline)
– Leading Edge Top View

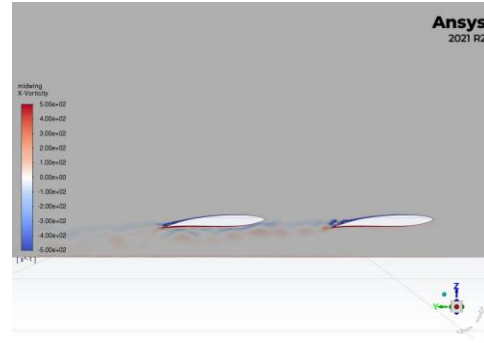


Figure 46: Instantaneous Vorticity Contour (Baseline)
– Midspan Side View

The top and midspan views (Figures 45 and 46) further show a rhythmic development and shedding of vortical structures downstream of the front wing. These vortices flow towards the rear wing in a periodic manner, indicating the presence of coherent wake-induced unsteadiness that governs the fluctuating aerodynamic loading experienced by the rear wing.

3.2.1. Effect of Endplate Gurney Flap on Time-Averaged Aerodynamic Performance

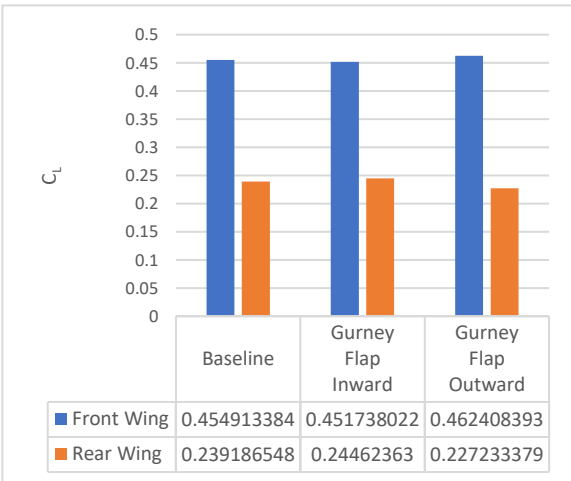


Figure 47: Time-Averaged C_L Comparison for Baseline and Gurney Flap Configurations (LES)

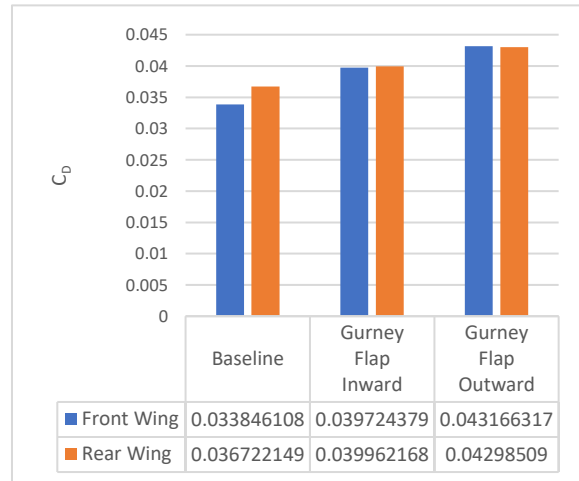


Figure 48: Time-Averaged C_D Comparison for Baseline and Gurney Flap Configurations (LES)

The introduction of the Gurney flap results in a modest redistribution of aerodynamic loading. For the front wing, the lift coefficient remains largely unchanged for the inward configuration and increases slightly for the outward configuration by approximately 2%, indicating minimal impact on the primary lifting surface.

For the rear wing, the inward-facing Gurney flap leads to a marginal increase in lift of about 2%, while the outward-facing configuration results in a decrease of approximately 5%, suggesting that the orientation of the flap significantly influences the wake interaction.

In terms of drag, both configurations exhibit an increase for the front and rear wings. The inward configuration produces a moderate drag increase of approximately 8% on front wing and 14% on rear wing, whereas the outward configuration results in a larger increase of around 22% on the front wing and 15% on the rear wing.

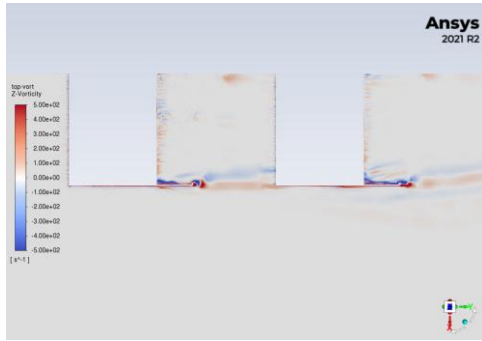


Figure 49: Instantaneous Vorticity Contour (Gurney Inward) – Leading Edge Top View

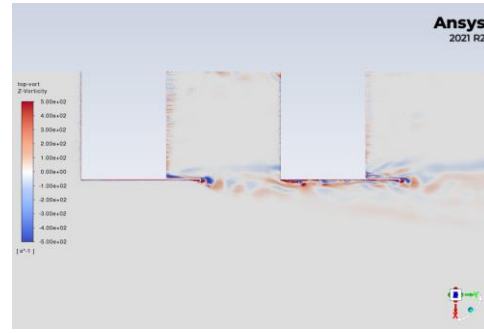


Figure 50: Instantaneous Vorticity Contour (Gurney Outward) – Leading Edge Top View

Overall, the Gurney flap modifies the wake structure and aerodynamic interaction between the wings, with the inward configuration offering a more balanced trade-off between lift and drag, while the outward configuration introduces higher aerodynamic penalties.

3.3. Spectral Analysis of Wake-Induced Loading

Fast Fourier Transform (FFT) analysis was performed [9] using the Python library `scipy.signal` on the time-resolved lift coefficient after subtraction of the mean value. The processed signal, after trimming to remove initial transient effects, consisted of 2001 uniformly sampled data points. To enable efficient FFT computation, the signal was zero-padded with additional samples to obtain a total of $N = 2048$ points.

For a discrete-time signal $x[n]$, where $n = 0, 1, 2, \dots, N - 1$, the Discrete Fourier Transform (DFT) is defined as

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-i2\pi \frac{kn}{N}}, k = 0, 1, 2, \dots, N - 1 \quad (12)$$

where $X[k]$ denotes the complex frequency-domain coefficients. With time step of $\Delta t = 0.001\text{s}$, corresponding to $N = 2048$ samples, the frequency sampling (f_s) and resolution (Δf) is given by

$$f_s = \frac{1}{\Delta t}, \Delta f = \frac{f_s}{N} \quad (13)$$

The one-sided power spectral density (PSD[k]) is obtained from the Fourier coefficients as

$$PSD[k] = \frac{2|X[k]|^2}{N f_s} \quad (14)$$

Welch method with Hanning window was employed to mitigate spectral leakage from rectangular window and variance in the spectral estimate due to the finite-time sampling of the simulation data [10, 11].

The Hanning window applied to each data segment is defined as

$$w[n] = \frac{1}{2} \left(1 - \cos \left(\frac{2\pi n}{N-1} \right) \right), n = 0, 1, 2, \dots, N-1 \quad (15)$$

The windowed signal is given by

$$x_w[n] = x[n] w[n] \quad (16)$$

For each segment, the DFT is computed to obtain $X_m[k]$, where m denotes the segment index. The Welch estimate of the power spectral density [1] is then calculated as

$$P_{xx}(k) = \frac{1}{M U f_s} \sum_{m=1}^M |X_m[k]|^2, U = \frac{1}{N} \sum_{n=0}^{N-1} w^2[n] \quad (17)$$

- M : number of overlapping segments
- U : window normalization factor

The power spectral density (PSD) of the aerodynamic coefficients was evaluated using Welch's method with a segment length of $N_{perseg} = 256$. This selection represents a deliberate trade-off between frequency resolution and statistical convergence. As demonstrated in the spectral convergence study, smaller segment sizes reduce variance through increased averaging, resulting in smoother spectra, while still preserving the dominant energy distribution across frequencies.

The chosen configuration therefore provides a statistically stable estimate with minimal spectral jaggedness, ensuring reliable identification of dominant flow-induced frequencies. (For complete plots refer to Appendix F)

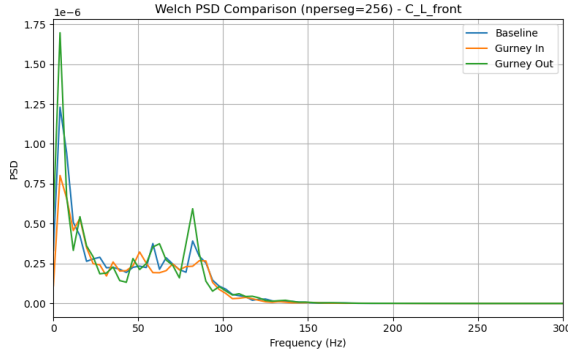


Figure 51: Spectral Analysis of Front Wing C_L

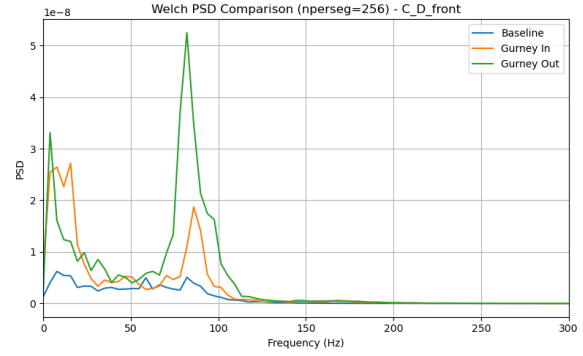


Figure 52: Spectral Analysis of Front Wing C_D

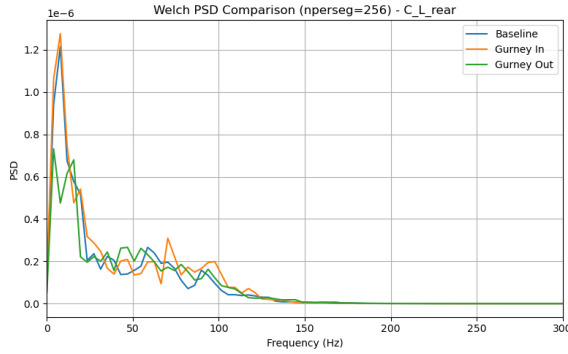


Figure 53: Spectral Analysis of Rear Wing C_L



Figure 54: Spectral Analysis of Rear Wing C_D

		Peak	Frequency [Hz]		
			Baseline	Gurney Inward	Gurney Outward
Front Wing	C_L	1	3.91	3.91	3.91
		2	7.81	7.81	7.81
		3	11.72	15.62	82.03
	C_D	1	7.81	15.62	82.03
		2	11.72	7.81	78.12
		3	15.62	3.91	85.94
Rear Wing	C_L	1	7.81	7.81	3.91
		2	3.91	3.91	15.62
		3	11.72	11.72	11.72
	C_D	1	7.81	7.81	3.91
		2	15.62	11.72	7.81
		3	11.72	15.62	11.72

Table 9: Frequency Peaks Comparison Between Configurations

		Peak	PSD [Unit ² /Hz]		
			Baseline	Gurney Inward	Gurney Outward
Front Wing	C_L	1	1.229966e-06	8.020761e-07	1.696507e-06
		2	9.439062e-07	6.588550e-07	6.675985e-07
		3	5.077148e-07	5.334859e-07	5.934182e-07
	C_D	1	6.220632e-09	2.719682e-08	5.248481e-08
		2	5.421936e-09	2.645229e-08	3.692057e-08
		3	5.352662e-09	2.540411e-08	3.474415e-08
Rear Wing	C_L	1	1.215226e-06	1.276006e-06	7.314073e-07
		2	9.409982e-07	1.061988e-06	6.793680e-07
		3	6.747354e-07	7.503796e-07	6.147912e-07
	C_D	1	4.948127e-09	1.755780e-08	2.493627e-08
		2	4.733429e-09	1.658066e-08	1.916764e-08
		3	4.507993e-09	1.593996e-08	1.735160e-08

Table 10: Power Spectral Density Peaks Comparison Between Configurations

The spectral results demonstrate clear differences in both energy magnitude and frequency distribution across the three configurations.

For the front wing C_L , the Gurney outward configuration produces the highest magnitude at the primary peak (3.91 Hz). However, unlike the baseline and Gurney inward cases, which remain confined below 20 Hz, the outward configuration introduces an additional high-frequency peak at 82.03 Hz. This indicates that part of the aerodynamic energy is redistributed from the dominant low-frequency mode to higher-frequency content. In contrast, the Gurney inward configuration shows lower overall PSD magnitude than the baseline across the dominant peaks, indicating a reduction in lift fluctuation energy without introducing new high-frequency components.

This behaviour becomes more pronounced in the front wing C_D . The Gurney outward case shows its dominant peak at 82.03 Hz, with the highest PSD magnitude concentrated at this frequency. The Gurney inward configuration also exhibits energy extension toward higher frequencies, although with lower magnitude compared to the outward case. In both configurations, the total spectral energy is higher than the baseline, as evidenced by the visibly larger area under the PSD curves. This confirms that the addition of a Gurney flap, regardless of orientation, increases the overall drag-related unsteady energy.

For the rear wing C_L , the effect differs between configurations. The Gurney outward case shows a reduction in energy at the dominant low-frequency peaks (3.91–11.72 Hz) without the appearance of new high-frequency peaks. This indicates a suppression of coherent low-frequency loading rather than redistribution. In contrast, both the baseline and Gurney inward configurations maintain their energy distribution within the same low-frequency band, with similar dominant frequencies, indicating that the wake interaction remains governed by large-scale structures.

For the rear wing C_D , both Gurney inward and Gurney outward configurations exhibit a clear redistribution of energy toward higher frequencies compared to the baseline. In addition, both configurations show significantly larger total spectral energy across frequency range, confirming an overall increase in drag-related unsteady loading. The baseline case remains the lowest in both magnitude and frequency spread.

Overall, the addition of a Gurney flap leads to a redistribution of spectral energy toward higher frequencies, particularly evident in drag. This behaviour is consistent with the generation of more energetic and fragmented flow structures. The increased presence of higher-frequency components suggests enhanced turbulent activity, indicating a shift toward more isotropic turbulence in the downstream wake. However, this comes with a clear trade-off: a substantial increase in drag-related energy, as reflected by the higher total spectral energy and broader spectral distribution.

4. Conclusion

The results demonstrate that tandem WIG aerodynamics are fundamentally governed by unsteady wake interactions rather than purely steady flow behaviour. While steady simulations capture the mean redistribution of aerodynamic loads between the front and rear wings, they fail to represent the transient phenomena that dominate the downstream wing environment.

Unsteady simulations reveal that the rear wing is subjected to coherent, time-varying aerodynamic forcing arising from the upstream wake. Spectral analysis confirms the presence of dominant frequencies, indicating that the loading is structured and periodic rather than random. This behaviour has direct implications for longitudinal stability, as it introduces dynamic forcing that cannot be addressed through steady aerodynamic design alone.

The findings further demonstrate that wake characteristics in tandem WIG configurations are not fixed but can be actively altered through geometric modifications. The introduction of Gurney flaps at the front-wing endplates provides clear evidence that wake structure can be manipulated, as observed through the redistribution of spectral energy and the reduction in large-scale coherent features.

However, this modification introduces a trade-off, where improvements in wake homogenization are accompanied by increased overall unsteady energy and aerodynamic drag. This indicates that while passive devices can influence the nature of wake interaction, they do not inherently reduce the total aerodynamic disturbance experienced by the rear wing.

Nevertheless, the ability to alter the wake structure presents a meaningful design pathway. It suggests that future configurations can be systematically tuned to achieve a more favourable balance between wake coherence, unsteady loading, and aerodynamic efficiency. As such, wake manipulation should be considered a viable direction in future tandem WIG design, particularly when integrated with control strategies that can respond to the modified frequency content of the flow.

Overall, the study establishes that accurate characterization of tandem WIG systems requires resolving unsteady flow physics.

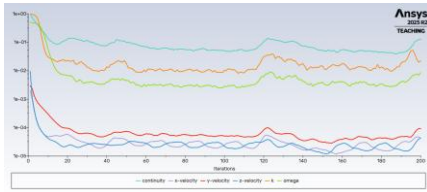
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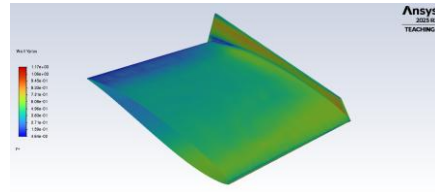
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Appendix A

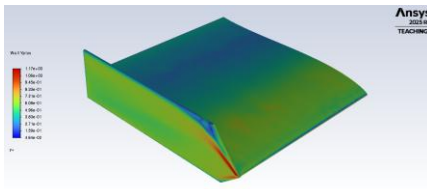
Mesh Convergence – Coarse



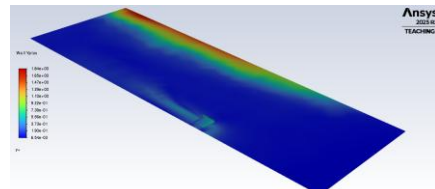
Residual Plot



Y^+ Contour Plot – Wing Top Surface

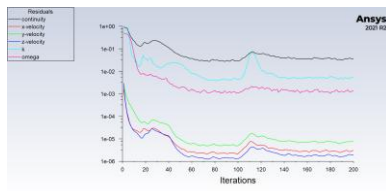


Y^+ Contour Plot – Wing Bottom Surface

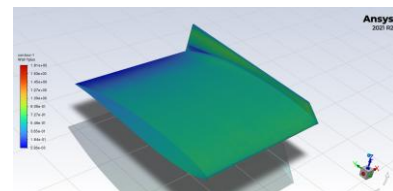


Y^+ Contour Plot – Wing and Ground

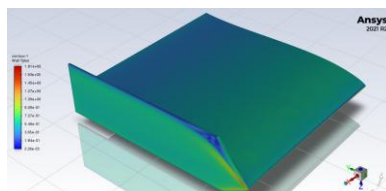
Mesh Convergence – Medium



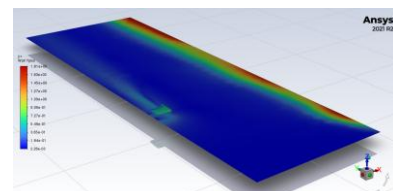
Residual Plot



Y^+ Contour Plot – Wing Top Surface

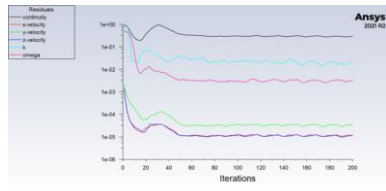


Y^+ Contour Plot – Wing Bottom Surface

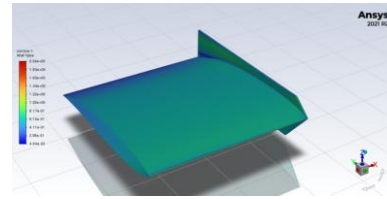


Y^+ Contour Plot – Wing and Ground

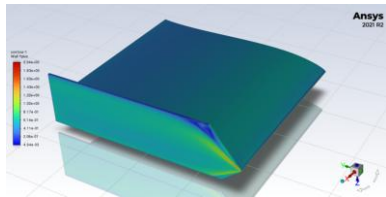
Mesh Convergence – Fine



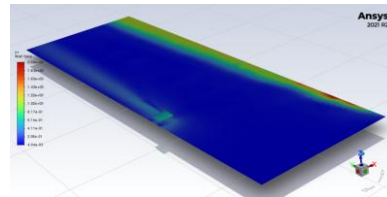
Residual Plot



Y^+ Contour Plot – Wing Top Surface

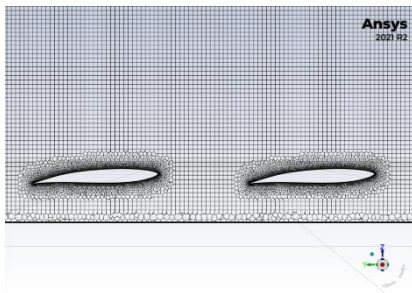


Y^+ Contour Plot – Wing Bottom Surface

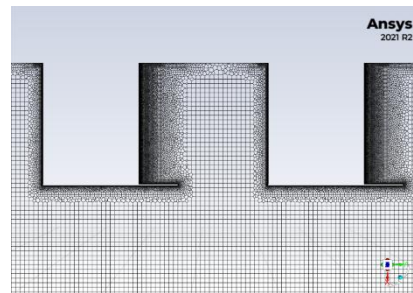


Y^+ Contour Plot – Wing and Ground

LES Mesh



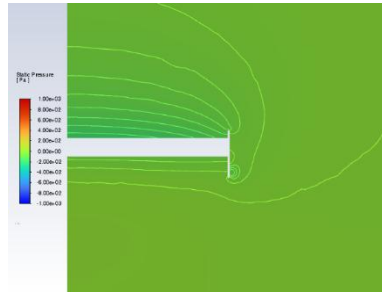
LES Mesh Zoomed – Midspan Side View



LES Mesh Zoomed – Leading Edge Top View

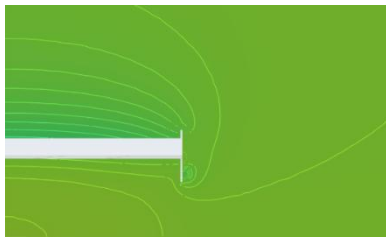
Appendix B

Single Wing OGE Pressure Contour Plots



$$\alpha = 5^\circ$$

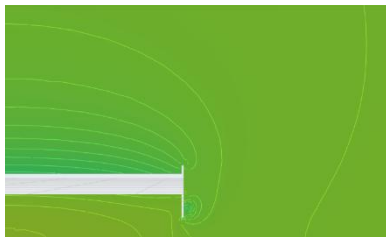
Single Wing IGE Pressure Contour Plots



$$h/b \text{ (Ashill)} = 0.245701303$$



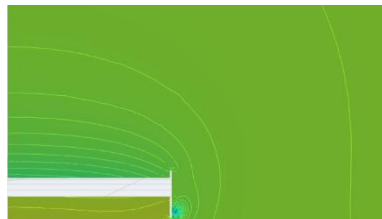
$$h/b \text{ (Ashill)} = 0.176576879$$



$$h/b \text{ (Ashill)} = 0.153535405$$

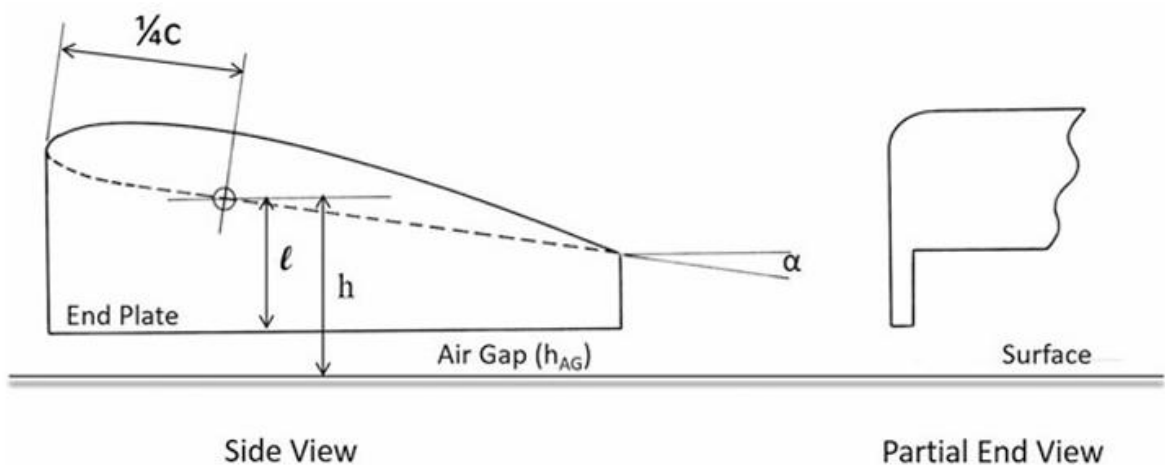


$$h/b \text{ (Ashill)} = 0.13049393$$



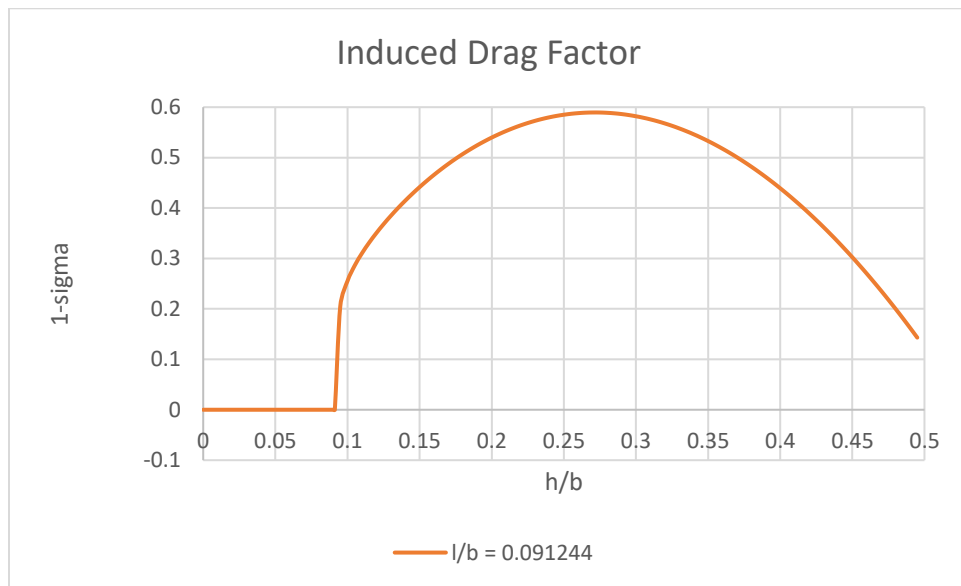
$$h/b \text{ (Ashill)} = 0.107452455$$

Appendix C



$$\frac{h_{AG}}{b} = \frac{h - \ell}{b}$$

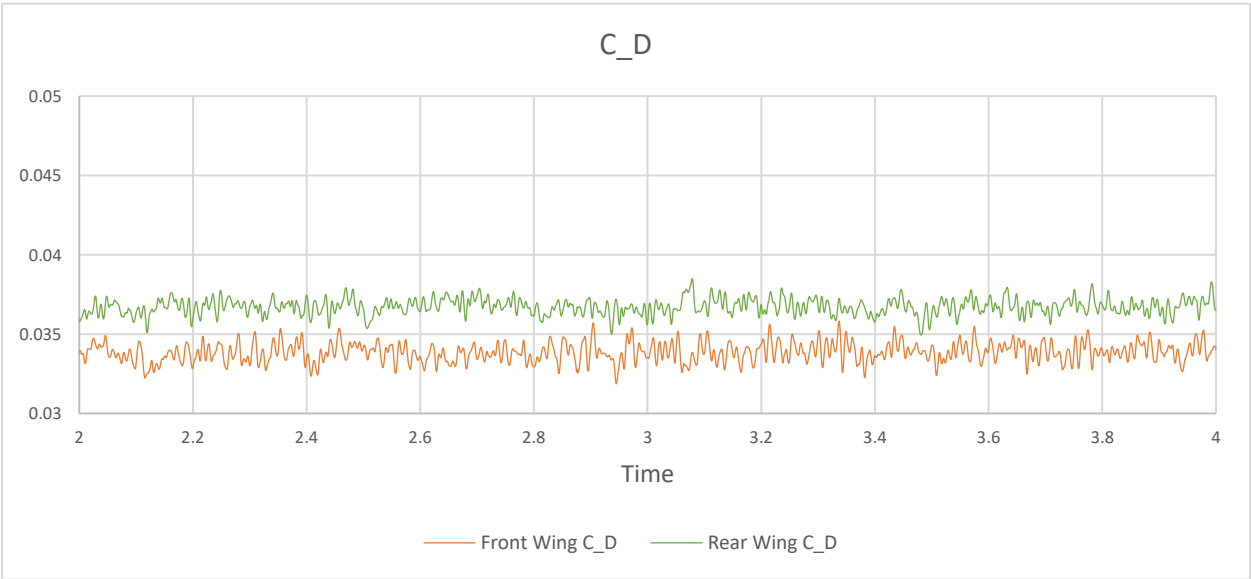
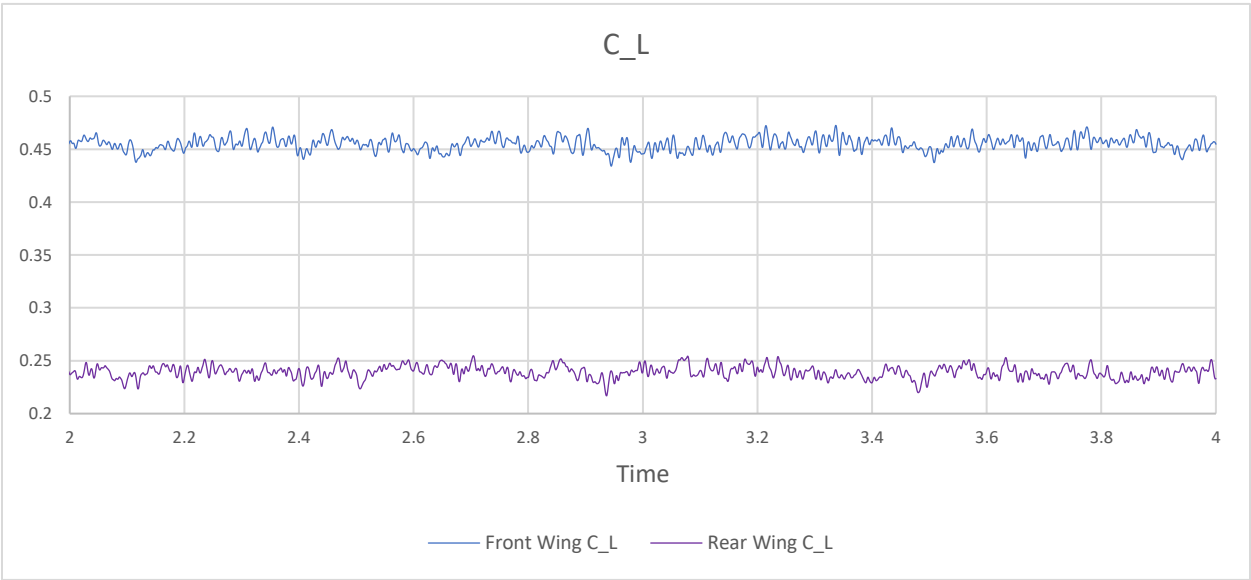
Dimension Nomenclature Diagram

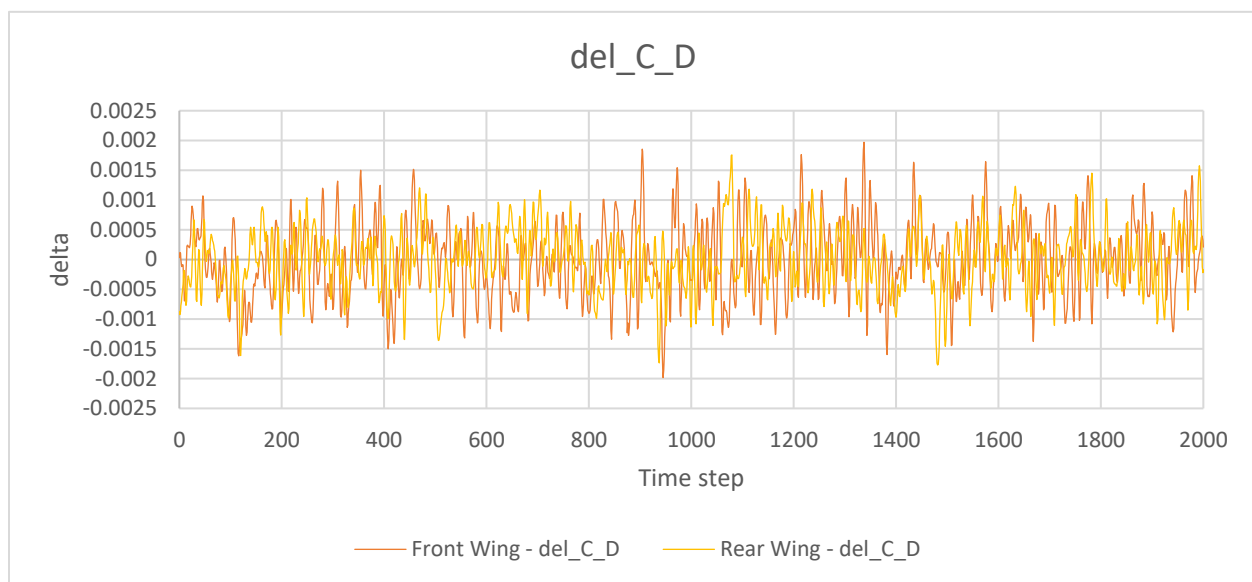
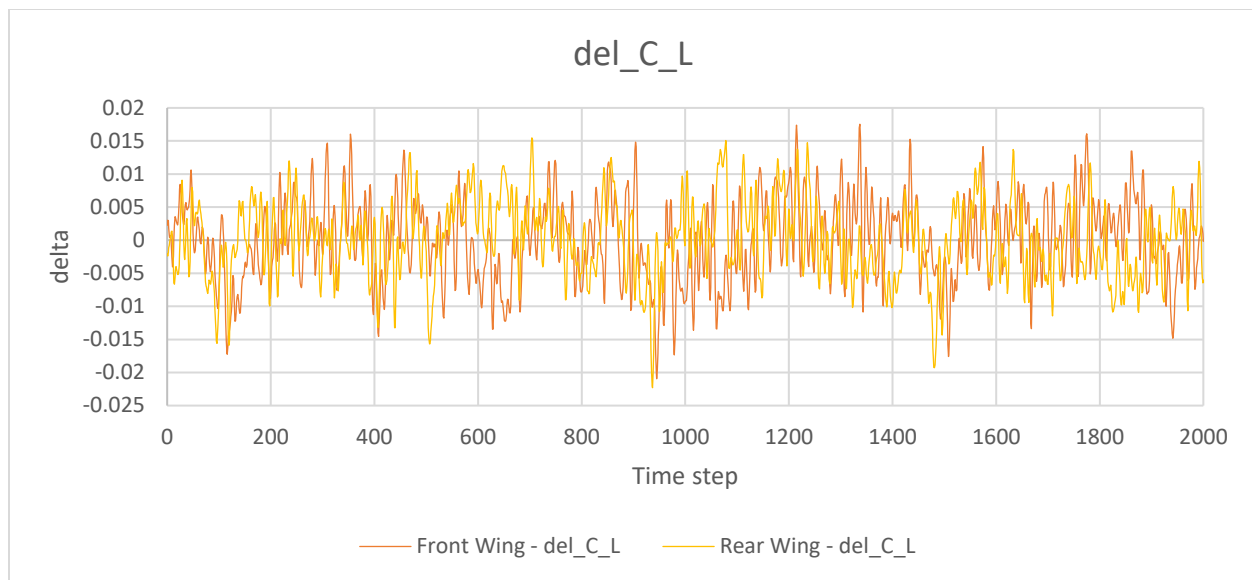


Ashill's Empirical Plot for $l/b=0.091244$

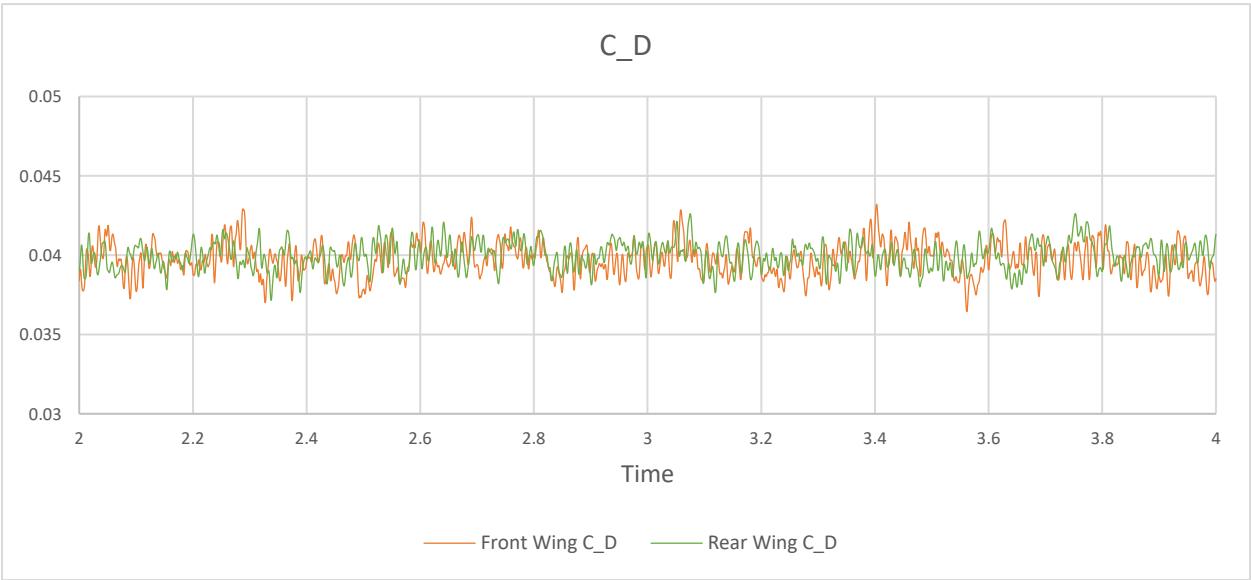
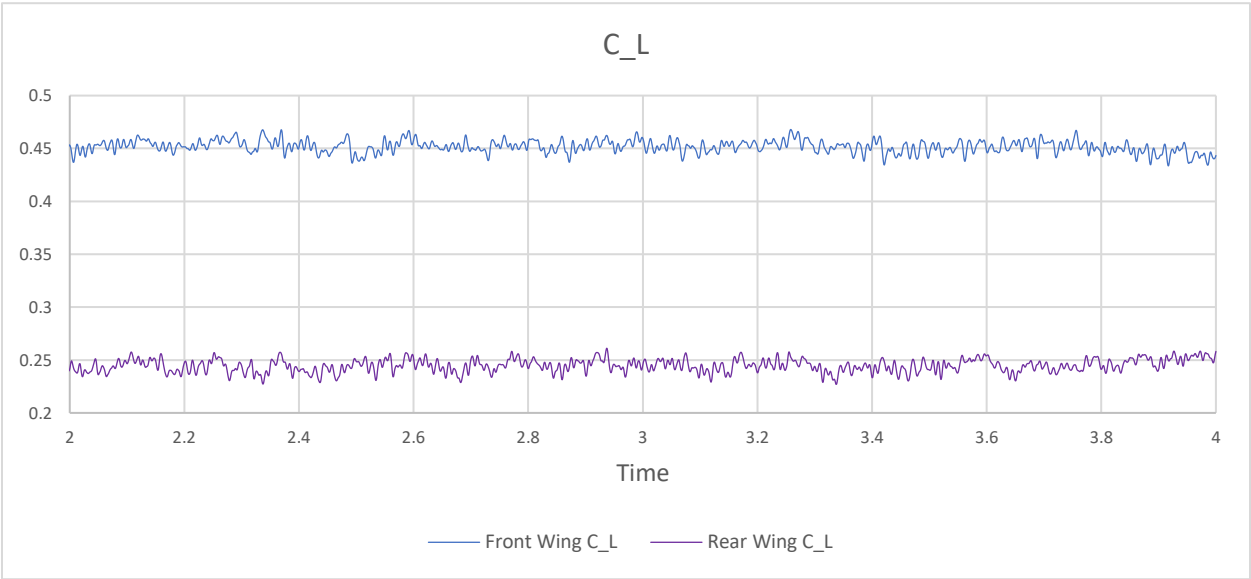
Appendix D

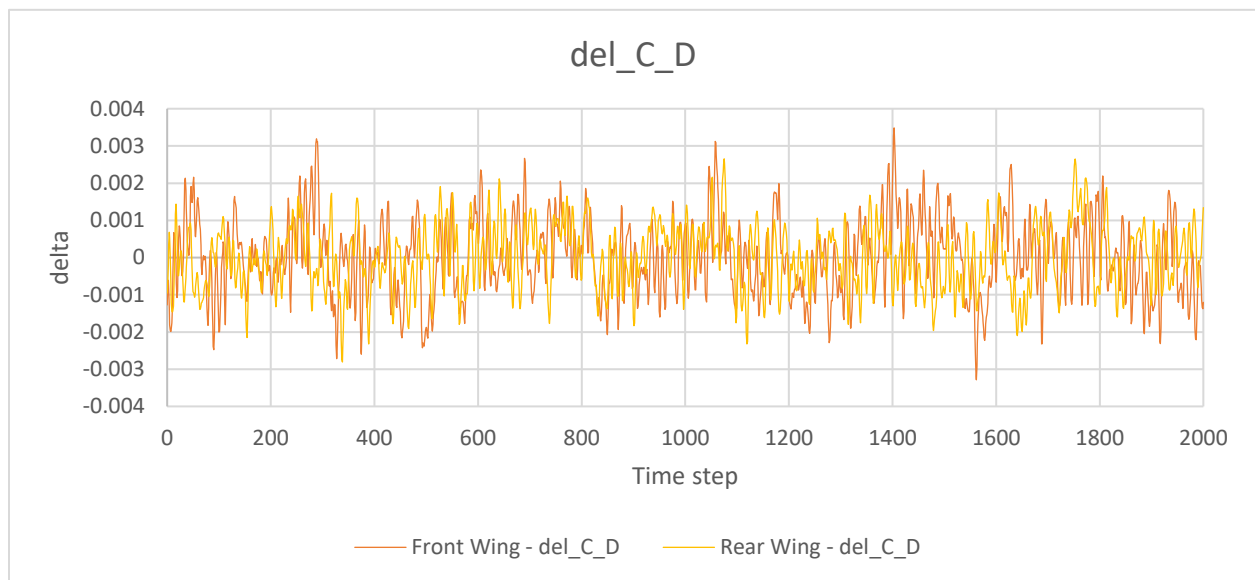
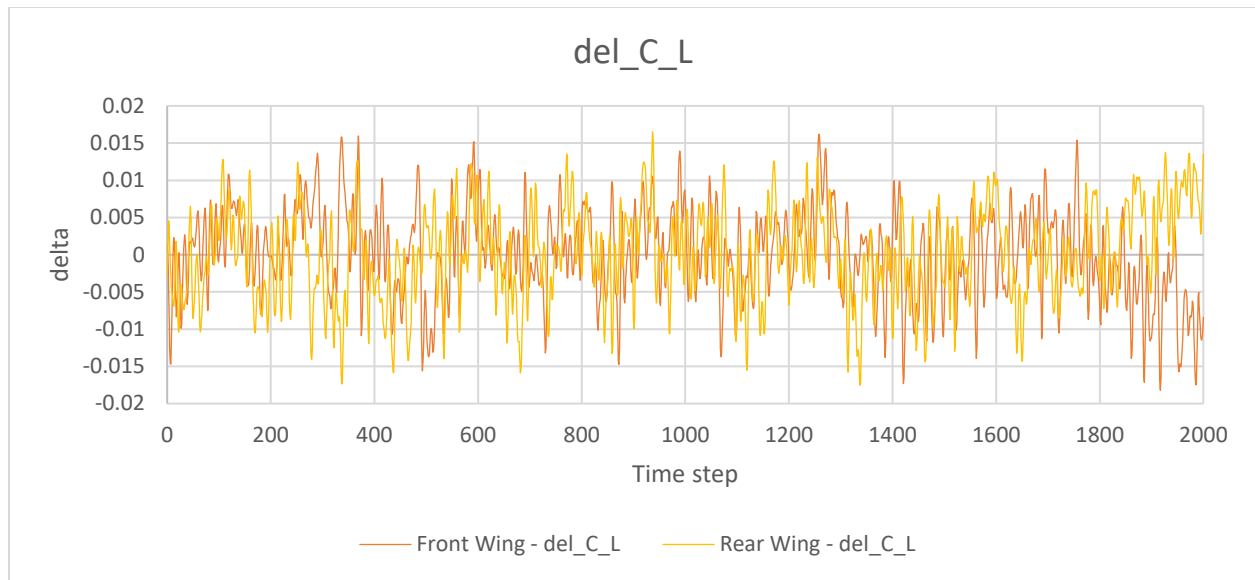
Baseline



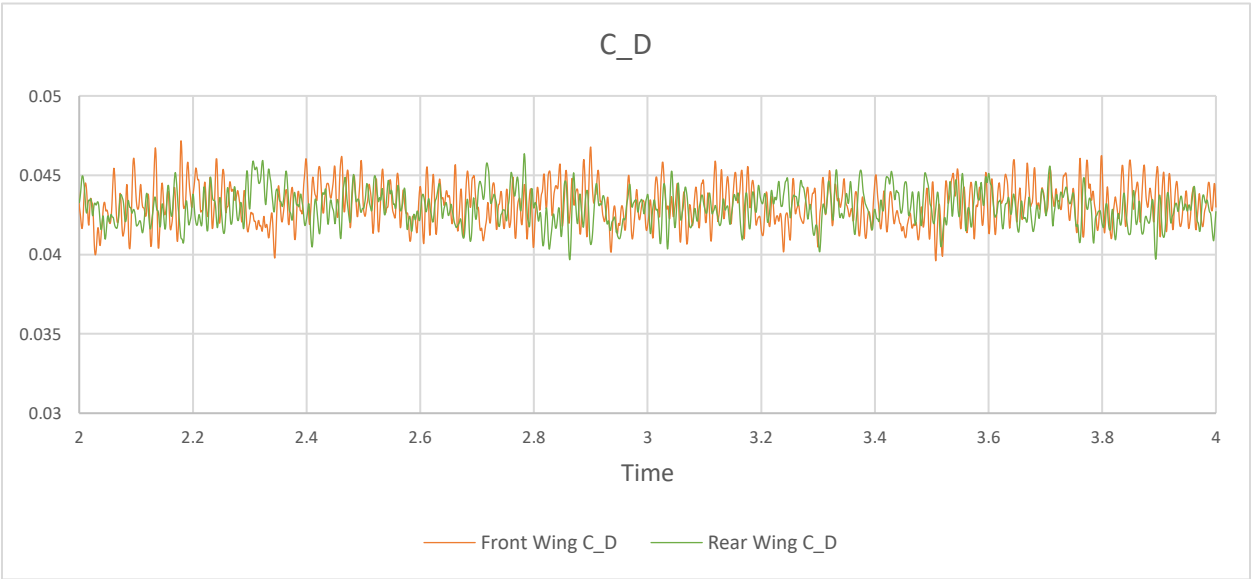
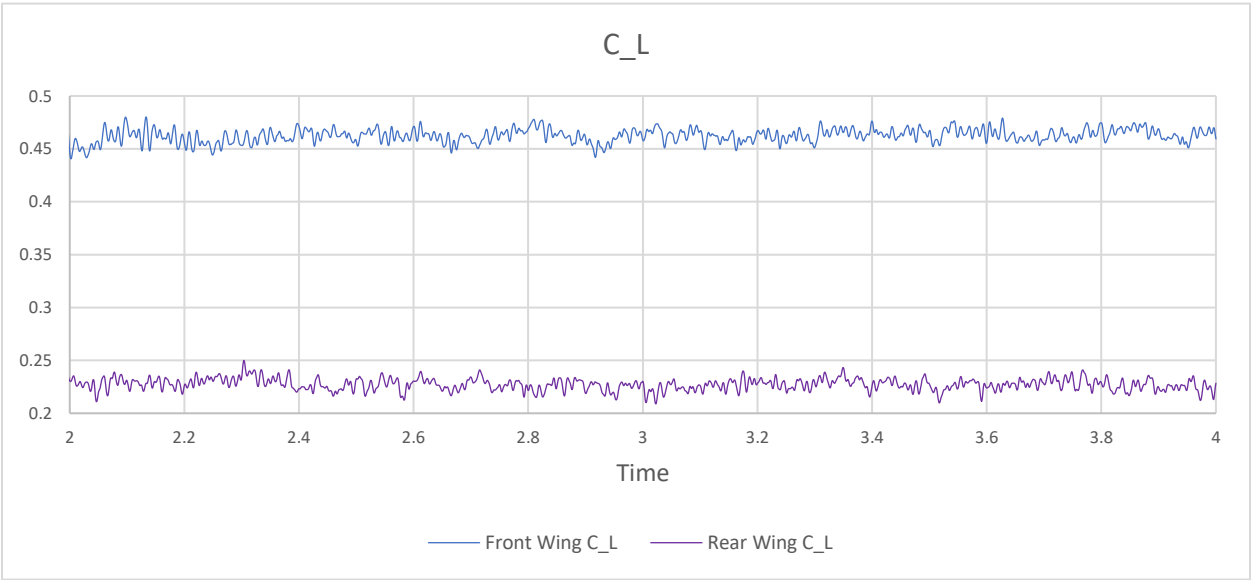


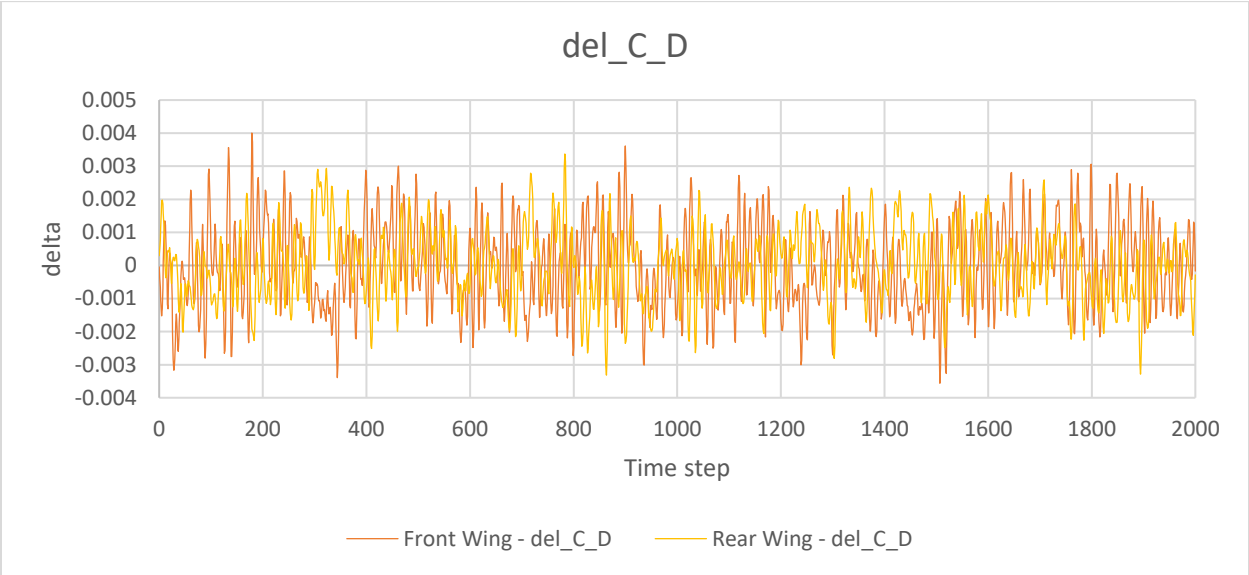
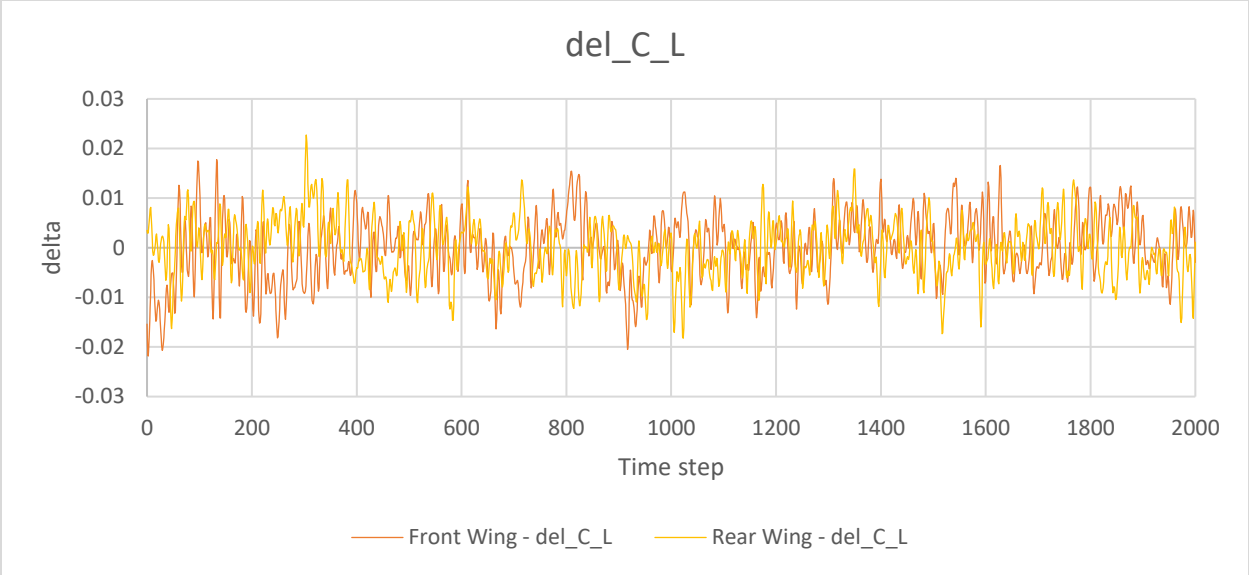
Gurney Flap Inward



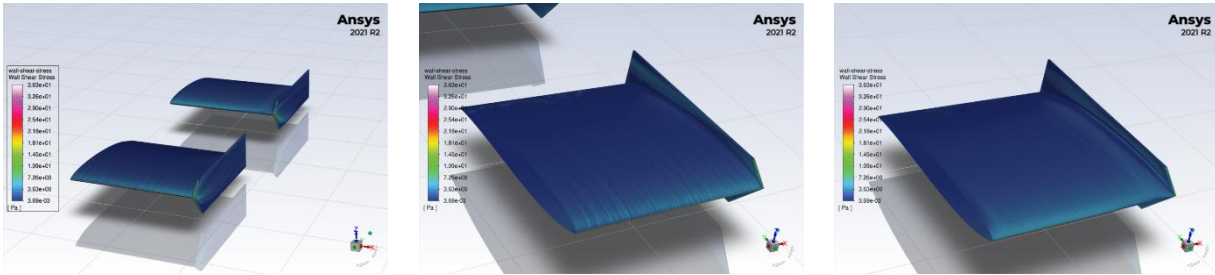


Gurney Flap Outward

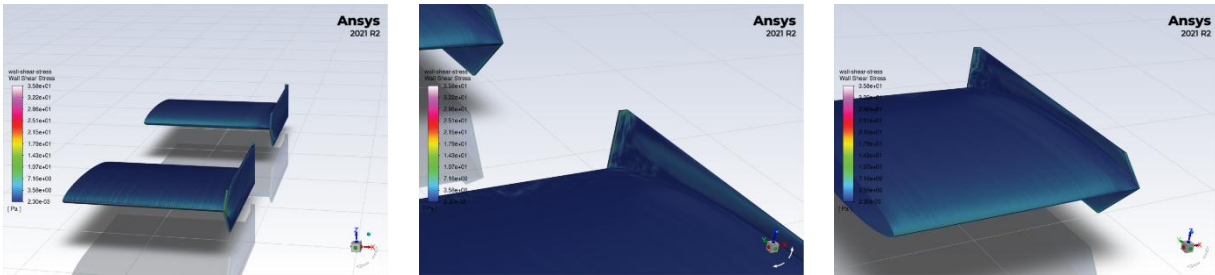




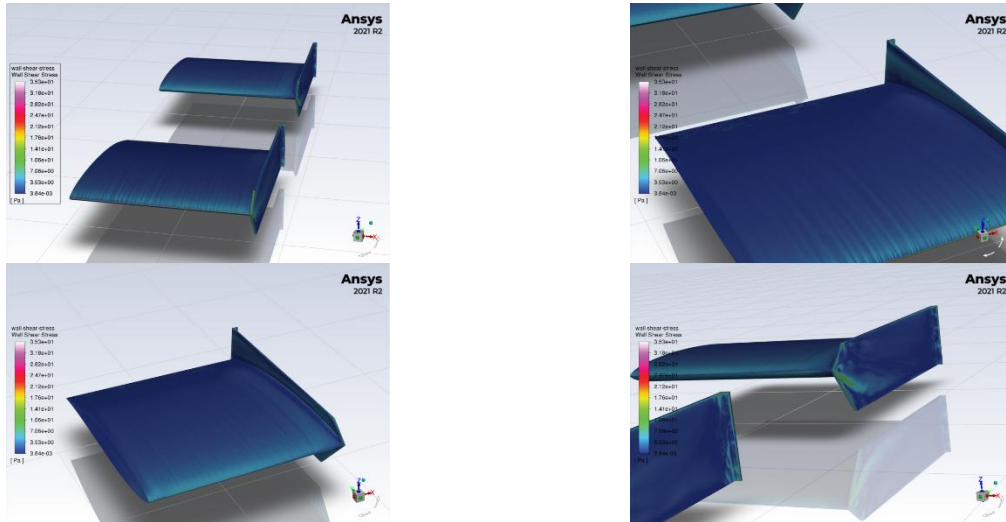
Appendix E



Instantaneous Wall Shear Stress Contour – Baseline



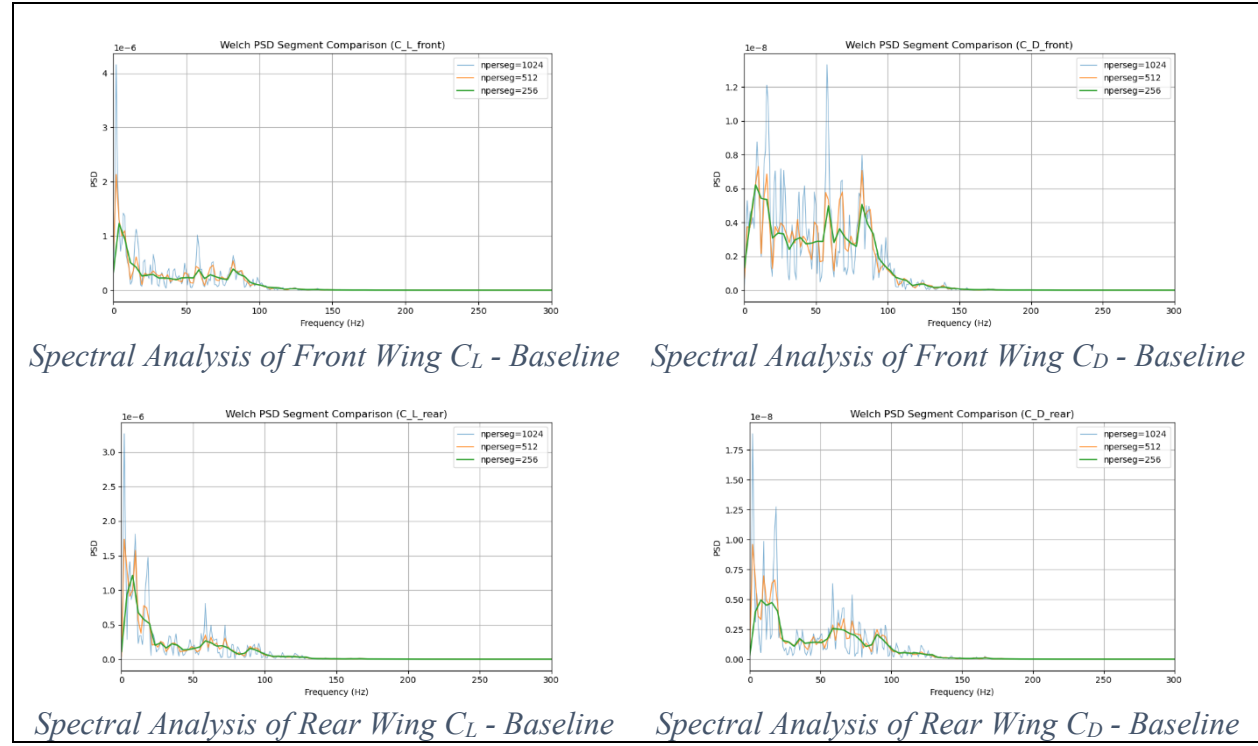
Instantaneous Wall Shear Stress Contour – Gurney Inward



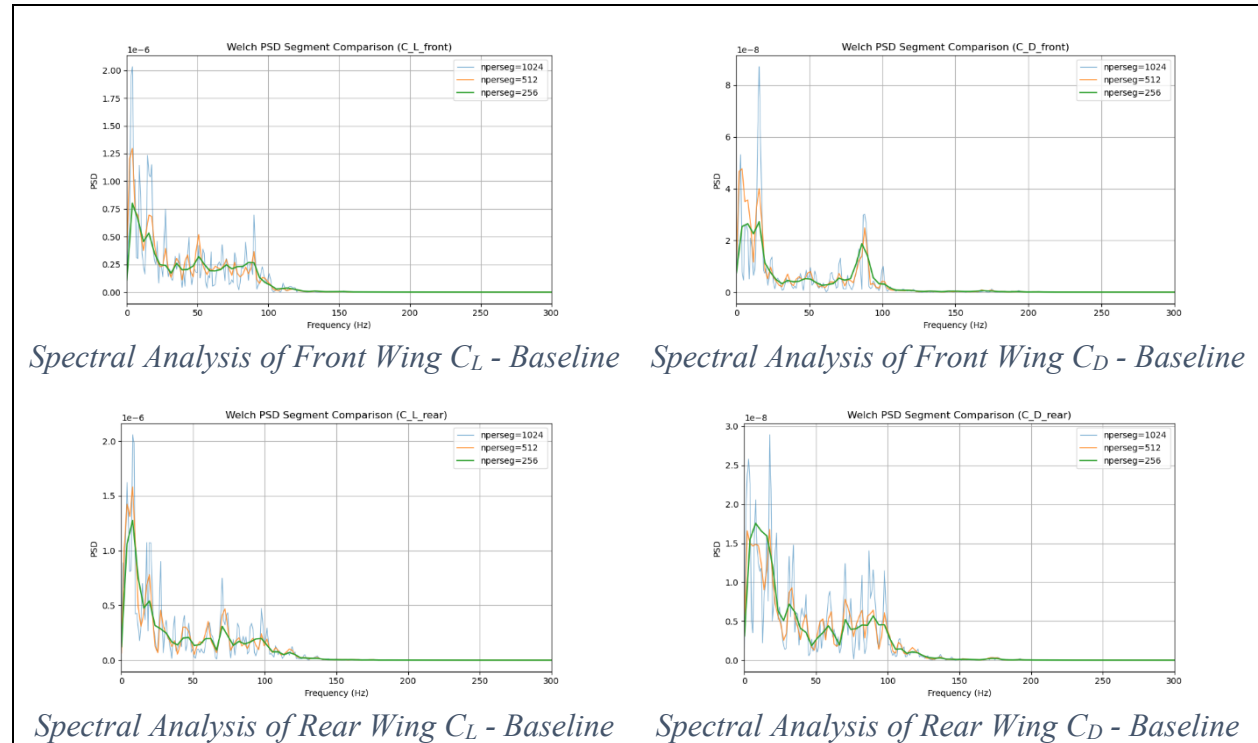
Instantaneous Wall Shear Stress Contour – Gurney Outward

Appendix F

Baseline



Gurney Inward



Gurney Outward

